Learning Precision-Selective Contact Skills from Behaviour Priors: Variability-Driven Variable Impedance Control with State-Corrected Task Parametrization and Torque-Port Energy Accounting

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Abstract

This report presents a simulated experimentation framework for testing the Thesis' approach for transferring a contact rich skill to a robot manipulator by learning from encoded human behaviour priors. The experimental setup represents a robot with a scalpel attached to its end effector learned to perform an adaptive cutting skill on variant types of materials. unlike replaying a single trajectory, the method learns the pattern of the skill and reproduces it under a variable-impedance controller with an implemented energy-tank mechanism at the joint torque port. The framework combines four elements. First, a *shared material-conditioned GMM/GMR* trained after spatial-progress registration. Second, a *variable impedance controller* whose stiffness matrix is learned from the consistency of the demonstrations: a quadratic program (QP) solved at each control step maps the demonstration precision onto the controller gains, so the manipulator is stiff only along directions that the demonstrations regulated and compliant elsewhere. The phase-domain pose, wrench and twist prior is shared across cork, penoplex and PVC; categorical material contrasts condition the common model without treating material identity as an ordinal scalar. Third, a *state-corrected task parametrisation* that indexes the motion by a progress (phase) variable whose evolution is corrected by the manipulator's measured state. Its incremental effect is evaluated rather than assumed. Fourth, sufficient stability conditions are derived for idealised error models, and the implemented torque projection is coupled to a discrete tank that is settled from delivered joint power. A positive-power perturbation experiment tests the resulting storage and power inequalities. These are simulation-level results, not a hardware-safety or human-interaction guarantee. An important consideration for the studied approach is the data repeatability to decide which skill patterns to extract under which constraints, thus, across repeated demonstrations position-channel correlation ranges from 0.37 to 0.98 and force-channel correlation from approximately 0.00 to 0.27 across the three datasets. The controller maps this continuous, axis-dependent precision to stiffness. Desired force enters the gain objective and a virtual-wrench bound; it is not a hard measured-contact-force constraint. A separate per-material *cutting-force model*, identified from each material's own data, provides the interaction force in simulation. The framework is evaluated with three separately calibrated materials (cork, PVC and penoplex), not by zero-shot generalisation to unseen materials.

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Nomenclature

The table below lists every symbol used, its meaning, and its physical dimension. Bold lower-case symbols are 3-vectors expressed in the world frame; n is the number of robot joints ($n = 7$ for the KUKA iiwa 14 used here).

Symbol	Meaning	Units
ϕ	task phase (progress along the skill), 0 at start, 1 at end	dimensionless
$x,\ x_d$	actual / desired scalpel-tip (TCP) position	m
$\dot{x},\ \dot{x}_d$	actual / desired scalpel-tip velocity	$\mathrm{m\,s^{-1}}$
$\tilde{x} = x_d - x$	position tracking error	m
$\dot{\tilde{x}} = \dot{x}_d - \dot{x}$	velocity tracking error	$\mathrm{m\,s^{-1}}$
f	commanded Cartesian force (impedance "virtual force")	N
f_{ext}	external (environment/contact) force on the scalpel	N
$K(\phi)$	diagonal stiffness matrix (per axis)	$\mathrm{N\,m^{-1}}$
$D(\phi)$	diagonal damping matrix	$\mathrm{N\,s\,m^{-1}}$
K_I	integral gain	$\mathrm{N\,m^{-1}s^{-1}}$
ζ	damping ratio (set to 0.7)	dimensionless
$\Lambda,\ M$	task-space (Cartesian) inertia	kg
$C\dot{q},\ g$	Coriolis/centrifugal and gravity joint torques	N m

Symbol	Meaning	Units
τ	commanded joint torques	$\mathrm{N\,m}$
$q, \dot{q}$	joint angles / velocities	rad, $\mathrm{rad\,s^{-1}}$
J	TCP Jacobian ($\dot{x} = J\dot{q}$), $3 \times n$	$\mathrm{m\,rad^{-1}}$
λ^2	damped-least-squares regulariser	$\mathrm{m^2\,rad^{-2}}$
N	null-space projector ($I - J^+J$)	dimensionless
$N_{\mathrm{seed}}, N_{\mathrm{demo},*}, N_{\mathrm{mat}}$	numbers of random seeds, demonstrations, and material-identification samples	dimensionless
$\sigma_i^2(\phi)$	cross-trial variance of (range-normalised) signal, axis i	dimensionless
$\Gamma_\chi, \mathcal{R}_\chi$	shape-correlation and regulation scores for signal channel χ	dimensionless
B_χ, U_χ	between-phase and within-phase variances of channel χ	channel units squared
$N_\phi, N_{\mathrm{boot}}$	phase samples per trial and bootstrap resamples	dimensionless
$\epsilon_\sigma, \epsilon_\tau$	precision and torque-projection numerical regularisers	dimensionless, $\mathrm{rad^2\,s^{-2}}$
$w_i(\phi)$	precision weight (normalised inverse variance) $\in [0, 1]$	dimensionless
$K_i^{\mathrm{pos}}(\phi)$	precision-derived position stiffness, axis i	$\mathrm{N\,m^{-1}}$
$K_{\mathrm{floor}}, K_{\mathrm{min}}, K_{\mathrm{max}}$	stiffness floor / box limits	$\mathrm{N\,m^{-1}}$
u, u_w	normalised candidate / identified stiffness allocations	dimensionless
$H(\phi)$	phase-dependent task-relevance matrix in the allocation proof	dimensionless
$q_f(\phi)$	force-consistency weight in the QP $\in [0, 1]$	dimensionless
F_d	desired interaction force (from the prior)	N
F_i^{bound}	per-axis demonstrated force limit	N
$\hat{\phi}$	state-estimated phase	dimensionless
$\phi^\star, \phi_c, e_\phi$	state-consistent phase, clock phase, and phase synchronisation error	dimensionless
$\dot{\phi}_{\mathrm{nom}}$	nominal (clock) phase rate	$\mathrm{s^{-1}}$
$\Delta\phi_k^{\mathrm{nom}}$	nominal phase increment in one controller sample	dimensionless
s_ϕ, T_d	execution-speed multiplier and demonstration duration	dimensionless, s
k_ϕ	phase-synchronisation rate	$\mathrm{s^{-1}}$
ρ_x, ρ_F	normalising scales for position / force matching	m, N
ℓ_ϕ^-, ℓ_ϕ^+	backward/forward phase-search horizons	dimensionless
$\gamma_{T,k}$	energy-tank phase gate at sample k, in $[0, 1]$	dimensionless
$\mathcal{C}_k$	discrete local candidate set used for phase estimation	dimensionless
$z, z_r, \mathcal{M}, W$	measured/reference task state, task manifold, and observation weighting	mixed
$\hat{t}_r, F_h, f_h$	unit path tangent, holdback magnitude, and validation force	dimensionless, N, N
$\mathcal{S}$	total passivity storage (mechanical energy plus tank)	J
V	stability Lyapunov function	J
T	energy-tank state	J
h_c, h_{sim}	controller and simulator integration periods	s

Symbol	Meaning	Units
P_{inj}	power drawn by time-varying gains / moving reference	W
β	Lyapunov cross-term coefficient	s^{-1}
α	nominal exponential decay rate	s^{-1}
A	nominal stability state matrix	mixed state-space units
P, Q	Lyapunov and dissipation matrices in the stability proof	mixed energy units
d	blade penetration depth below the material surface	m
k_{mat}	material loading stiffness	$\text{N}\,\text{m}^{-1}$
$F_{\text{mat}}^{\text{cut}}$	material cutting-force plateau	N

1 Problem statement

The robot's task is to acquire a contact-rich manipulation skill, cutting, from a handful of human encoded behaviour priors. The behaviour priors are provided to the robot controller from behaviour prior probabilistic generator trained to encode human demonstrated position, twist and wrench profiles as input to the robot control algorithm. Each demonstration provides, sampled over time, the scalpel pose, its twist (linear/angular velocity), and the interaction wrench measured at the scalpel. This aims to capture the *skill* instead of cloning the behaviour. i.e., a controller that executes the task with quantified error, respects the demonstrated variability structure and remains bounded in the tested simulation conditions.

The position and force dependency during contact through the motion path represents the core challenge in the study. In a direction where the scalpel presses on the material, fixing the position determines the force and vice-versa; commanding robot controller to track both is non trivial issue. A principled controller must therefore *decide what to regulate in each direction*, and a reliable simulation must *model the material* that gives rise to the contact force in the first place. Both points are addressed.

2 Patterns in the demonstration that represent a reliable skill

The reliable skill patterns were extracted from data repeatability across trails. i.e, the repeatable components of the demonstration to be treated as the skill where other components are constraints to shape the behaviour of performing the skill by robot. For each signal, across trials measurements of (a) the *shape correlation* and (b) a *regulation score* were reported. shape correlation measures how similarly the signal evolves from trial to trial after removing its mean where *regulation score* refers to the fraction of the signal's variance that is explained by the task phase, i.e. a phase-locked signal-to-noise ratio.

Let $y_{\nu,\chi}(\phi_\ell)$ denote channel χ in demonstration $\nu \in \{1, \ldots, N_{\text{demo}}\}$ after all trials have been resampled at the common phase samples ϕ_ℓ, $\ell \in \{1, \ldots, N_\phi\}$, with $N_\phi = 300$. The trial-specific temporal mean and demeaned profile are

$$\bar{y}_{\nu,\chi}^{\phi} = \frac{1}{N_\phi} \sum_{\ell=1}^{N_\phi} y_{\nu,\chi}(\phi_\ell), \qquad \breve{y}_{\nu,\chi}(\phi_\ell) = y_{\nu,\chi}(\phi_\ell) - \bar{y}_{\nu,\chi}^{\phi}. \tag{1}$$

Here ν indexes demonstrations, χ identifies one of X, Y, Z, F_x, F_y, F_z, and ℓ indexes phase. Subtracting $\bar{y}_{\nu,\chi}^{\phi}$ removes constant trial-dependent offsets before shape comparison. The pairwise

shape correlation is

$$\mathrm{corr}_{\nu\mu}^{(\chi)} = \frac{\sum_{\ell=1}^{N_\phi} \breve{y}_{\nu,\chi}(\phi_\ell)\breve{y}_{\mu,\chi}(\phi_\ell)}{\sqrt{\sum_{\ell=1}^{N_\phi} \breve{y}_{\nu,\chi}^2(\phi_\ell)}\sqrt{\sum_{\ell=1}^{N_\phi} \breve{y}_{\mu,\chi}^2(\phi_\ell)}}, \qquad \nu < \mu, \tag{2}$$

and the reported channel statistic is the mean over all distinct trial pairs,

$$\Gamma_\chi = \frac{2}{N_{\mathrm{demo}}(N_{\mathrm{demo}}-1)} \sum_{\nu<\mu} \mathrm{corr}_{\nu\mu}^{(\chi)}\,. \tag{3}$$

Thus $\Gamma_\chi \in [-1, 1]$ measures reproducibility of the evolving profile, not agreement of its absolute offset.

The phase-conditioned mean and grand mean are

$$\bar{\breve{y}}_\chi(\phi_\ell) = \frac{1}{N_{\mathrm{demo}}} \sum_{\nu=1}^{N_{\mathrm{demo}}} \breve{y}_{\nu,\chi}(\phi_\ell), \qquad \bar{\breve{y}}_\chi^{\mathrm{all}} = \frac{1}{N_\phi} \sum_{\ell=1}^{N_\phi} \bar{\breve{y}}_\chi(\phi_\ell). \tag{4}$$

The between-phase and within-phase variances are then

$$B_\chi = \frac{1}{N_\phi} \sum_{\ell=1}^{N_\phi} \big(\bar{\breve{y}}_\chi(\phi_\ell) - \bar{\breve{y}}_\chi^{\mathrm{all}}\big)^2, \tag{5}$$

$$U_\chi = \frac{1}{N_{\mathrm{demo}} N_\phi} \sum_{\nu=1}^{N_{\mathrm{demo}}} \sum_{\ell=1}^{N_\phi} \big(\breve{y}_{\nu,\chi}(\phi_\ell) - \bar{\breve{y}}_\chi(\phi_\ell)\big)^2. \tag{6}$$

B_χ quantifies the common trajectory variation explained by phase, while U_χ quantifies trial-to-trial variation remaining at the same phase. The regulation score is

$$\mathcal{R}_\chi = \frac{B_\chi}{B_\chi + U_\chi} \in [0, 1]. \tag{7}$$

A value near one indicates that the signal is predominantly phase locked; a value near zero indicates that knowing the phase does not predict the measured channel reliably across demonstrations.

Proposition 1 (Reliability interpretation of the two metrics)**.** *Suppose an aligned channel admits the repeated-trial decomposition*

$$y_{\nu,\chi}(\phi) = o_{\nu,\chi} + \psi_\chi(\phi) + \upsilon_{\nu,\chi}(\phi), \tag{8}$$

where $o_{\nu,\chi}$ is a trial-dependent constant offset, $\psi_\chi(\phi)$ is a common phase-locked pattern, and $\upsilon_{\nu,\chi}$ is zero-mean trial-specific variation independent across trials and uncorrelated with ψ_χ. If

$$\mathrm{Var}_\phi[\psi_\chi] = \varsigma_{\psi,\chi}^2, \qquad \mathrm{Var}_\phi[\upsilon_{\nu,\chi}] = \varsigma_{\upsilon,\chi}^2$$

is the same for all trials, then, in the large-sample limit,

$$\mathbb{E}\Big[\mathrm{corr}_{\nu\mu}^{(\chi)}\Big] = \frac{\varsigma_{\psi,\chi}^2}{\varsigma_{\psi,\chi}^2 + \varsigma_{\upsilon,\chi}^2}, \qquad \mathcal{R}_\chi \longrightarrow \frac{\varsigma_{\psi,\chi}^2}{\varsigma_{\psi,\chi}^2 + \varsigma_{\upsilon,\chi}^2}. \tag{9}$$

Consequently, both statistics estimate the phase-locked fraction of the dynamic signal and are invariant to the constant offset $o_{\nu,\chi}$.

Proof. Demeaning (8) by (1) eliminates $o_{\nu,\chi}$. For distinct trials $\nu \neq \mu$, independence of the residuals and their lack of correlation with ψ_χ give

$$\mathrm{Cov}_\phi(\breve{y}_{\nu,\chi}, \breve{y}_{\mu,\chi}) = \varsigma^2_{\psi,\chi},$$

whereas each demeaned trial has variance $\varsigma^2_{\psi,\chi} + \varsigma^2_{\upsilon,\chi}$. Division of the covariance by the two standard deviations yields the first expression in (9). For the variance decomposition, the phase-conditioned demeaned trial mean converges to the centred common pattern $\psi_\chi(\phi) - \mathbb{E}_\phi[\psi_\chi]$, so $B_\chi \to \varsigma^2_{\psi,\chi}$, while deviations about that mean give $U_\chi \to \varsigma^2_{\upsilon,\chi}$. Substitution into (7) yields the second expression. Neither limit depends on $o_{\nu,\chi}$. □

Quantity	cross-trial shape correlation	regulation score
Position (X/Y/Z)	0.37–0.98	0.42–0.95
Force (Fx/Fy/Fz)	−0.00–0.27	0.03–0.31

Table 2: How repeatable each quantity is across demonstrations (cork $N_{\mathrm{demo,cork}}$=28, PVC $N_{\mathrm{demo,PVC}}$=33, penoplex $N_{\mathrm{demo,peno}}$=22, 300 samples per trial).

The numerical separation between the two signal classes is

$$m_\Gamma = \min_{\substack{\text{material} \\ \chi \in \{X,Y,Z\}}} \Gamma_\chi - \max_{\substack{\text{material} \\ \chi \in \{F_x,F_y,F_z\}}} \Gamma_\chi = 0.373 - 0.266 = 0.107 > 0. \tag{10}$$

Thus even the least repeatable position channel has substantially greater shape agreement than the most repeatable force channel. Under Proposition 1, either reliability statistic $q_\chi \in \{\Gamma_\chi, \mathcal{R}_\chi\}$ corresponds to the phase-locked signal-to-residual ratio

$$\mathrm{SNR}_\chi = \frac{\varsigma^2_{\psi,\chi}}{\varsigma^2_{\upsilon,\chi}} = \frac{q_\chi}{1 - q_\chi}. \tag{11}$$

Across all materials and axes, position correlation is 0.373–0.983, whereas force correlation is −0.003–0.266. Regulation-score ranges are 0.418–0.947 for position and 0.032–0.310 for force. Position is therefore more repeatable in these datasets, but not every position direction is highly repeatable and force is not wholly unstructured.

Sampling uncertainty can be assessed without assuming normally distributed signals by resampling complete demonstrations. If $\Gamma^{*(b)}_\chi$ and $\mathcal{R}^{*(b)}_\chi$ are recomputed from bootstrap resample $b \in \{1, \ldots, N_{\mathrm{boot}}\}$, their percentile intervals are

$$\mathrm{CI}_{0.95}(\Gamma_\chi) = \left[Q_{0.025}\left(\Gamma^*_\chi\right), Q_{0.975}\left(\Gamma^*_\chi\right)\right], \qquad \mathrm{CI}_{0.95}(\mathcal{R}_\chi) = \left[Q_{0.025}\left(\mathcal{R}^*_\chi\right), Q_{0.975}\left(\mathcal{R}^*_\chi\right)\right], \tag{12}$$

where Q_p denotes the empirical p-quantile across bootstrap resamples. This trial-level resampling preserves the temporal dependence within each trajectory. A conservative reliability decision is obtained when the lower confidence limit of every regulated channel exceeds the upper confidence limit of every released channel. Equation (10) reports the observed effect-size margin; confidence intervals should accompany it when the bootstrap results are reported.

The defensible conclusion is that position is more phase repeatable than force in the analysed datasets, especially in the task-critical depth direction. The controller therefore maps the continuous per-axis precision to stiffness instead of imposing a binary position/force classification. The simulated interaction force is generated separately by a calibrated material model.

3 Variability-driven variable impedance control

3.1 The impedance control law

The robot is controlled as a virtual mechanical impedance at the scalpel tip. The commanded Cartesian force is

$$\boxed{f \;=\; K(\phi)\,\tilde{x} \;+\; K_I\xi \;+\; D(\phi)\,\dot{\tilde{x}}} \tag{13}$$

where,

- $K(\phi)\,\tilde{x}$ is a position-error spring. $K(\phi)\,[\mathrm{N\,m^{-1}}]$ is a diagonal stiffness (one value per Cartesian axis) and $\tilde{x}\,[\mathrm{m}]$ is the position error, so the product has units of force [N]. A larger K pulls the scalpel back to the reference more firmly.
- $K_I\xi$ compensates persistent contact-load error. The sampled accumulator ξ is clipped componentwise and uses conditional anti-windup.
- $D(\phi)\,\dot{\tilde{x}}$ is a velocity-error damper that suppresses oscillation; $D\,[\mathrm{N\,s\,m^{-1}}]$ times $\dot{\tilde{x}}\,[\mathrm{m\,s^{-1}}]$ gives [N].

The implemented integral update is

$$\xi_{k+1}^{\text{cand}} = \Pi_{[-\xi_{\max},\xi_{\max}]}(\xi_k + h_c\tilde{x}_k), \tag{14}$$

$$\xi_{i,k+1} = \begin{cases} \xi_{i,k}, & |f_{i,k}^{\text{cand}}| > F_i^{\text{bound}} \text{ and } \operatorname{sgn}(f_{i,k}^{\text{cand}}) = \operatorname{sgn}(\tilde{x}_{i,k}), \\ \xi_{i,k+1}^{\text{cand}}, & \text{otherwise}, \end{cases} \tag{15}$$

where $f^{\text{cand}} = K\tilde{x} + K_I\xi^{\text{cand}} + D\dot{\tilde{x}}$, $K_I = 800$ and $\xi_{\max} = 0.05\,\mathrm{m\,s}$. The residual wrench is then clipped componentwise.

The damping is tied to the stiffness by the heuristic $D(\phi) = 2\zeta\sqrt{M_{\text{eff}}K(\phi)}$, with $\zeta = 0.7$ and $M_{\text{eff}} = 2\,\mathrm{kg}$. This is not exact critical damping of the configuration-dependent Cartesian inertia. The virtual force is realised by joint torques through a damped-least-squares Jacobian transpose, with a posture task in the null space and model-based gravity/Coriolis compensation:

$$\tau \;=\; J^{\top}\big(JJ^{\top} + \lambda^2 I\big)^{-1} f \;+\; N\,\tau_{\text{posture}} \;+\; g + C\dot{q}, \qquad N = I - J^{\top}\big(JJ^{\top} + \lambda^2 I\big)^{-1} J. \tag{16}$$

where $J\,[\mathrm{m\,rad^{-1}}]$ is the scalpel Jacobian ($\dot{x} = J\dot{q}$); $\big(JJ^{\top} + \lambda^2 I\big)^{-1}$ is the damped pseudo-inverse that maps the Cartesian force to joint torques while remaining numerically robust near kinematic singularities (λ^2 is a small regulariser); N projects the secondary posture torque τ_{posture} into the null space so it does not disturb the task; and $g + C\dot{q}\,[\mathrm{N\,m}]$ cancel gravity and Coriolis effects so that (13) governs the closed loop.

3.2 Gains learned from demonstration precision

The *novice* work here is that $K(\phi)$ is not chosen manually, nor tabulated or tuned via deterministic regulation but it is the *precision* (inverse variance) of the demonstrations. Intuitively, where the human was very repeatable, that direction matters and should be tracked stiffly; where they were not, the controller should be soft. For each phase ϕ and axis i a precision weight on one shared scale was formed and mapped to stiffness. The calculation is made explicitly as follows. Let N_{demo} be the number of demonstrations and let $y_{\nu,i}(\phi)$ be the value of channel i in demonstration ν at phase ϕ. First, the mean across demonstrations at that same phase is

$$\bar{y}_i(\phi) = \frac{1}{N_{\text{demo}}} \sum_{\nu=1}^{N_{\text{demo}}} y_{\nu,i}(\phi). \tag{17}$$

The cross-trial standard deviation is then

$$s_i(\phi) = \sqrt{\frac{1}{N_{\text{demo}}} \sum_{\nu=1}^{N_{\text{demo}}} \left(y_{\nu,i}(\phi) - \bar{y}_i(\phi)\right)^2}. \tag{18}$$

Thus, $s_i(\phi)$ is small when all demonstrations have nearly the same channel value at phase ϕ, and it is large when their values are widely scattered. The denominator is N_{demo}, rather than $N_{\text{demo}} - 1$, matching the population-standard-deviation convention used in the implementation. Equation (18) is evaluated separately at each of the $N_\phi = 300$ phase samples for X, Y, Z, F_x, F_y, F_z, producing a 300×6 array.

Because the six raw channels have different units and numerical scales, their standard deviations cannot be compared directly. Each channel is therefore normalised by the range of its across-trial mean trajectory,

$$r_i = \max_\phi \bar{y}_i(\phi) - \min_\phi \bar{y}_i(\phi), \qquad \sigma_i(\phi) = \frac{s_i(\phi)}{\max(r_i, \epsilon_r)}, \tag{19}$$

where ϵ_r only protects against a zero range. The resulting $\sigma_i(\phi)$ is a dimensionless cross-trial inconsistency: it measures the scatter at that phase as a fraction of the total demonstrated motion or force range of that channel. Its inverse variance gives the precision

$$p_i(\phi) = \frac{1}{\sigma_i^2(\phi) + \epsilon_\sigma}. \tag{20}$$

Low relative scatter therefore gives high precision, while high relative scatter gives low precision.

To place every axis and channel on one common scale, all $6N_\phi = 1800$ precision values for one material are pooled into the set

$$\mathcal{P} = \{p_i(\phi_\ell) : i \in \{X, Y, Z, F_x, F_y, F_z\},\ \ell = 1, \ldots, N_\phi\}, \qquad p_{\text{ref}} = \text{percentile}_{95}(\mathcal{P}). \tag{21}$$

Thus a separate percentile is not calculated for each axis: the position and force precisions at all 300 phases contribute to the same p_{ref}. Approximately 95% of the pooled precision values are at or below this reference and the upper 5% are above it. The reference is shared across all six channels within a material, but it is recomputed from that material's own demonstrations. For the present data, $p_{\text{ref}} = 42.669$ for cork, 35.895 for PVC, and 18.734 for penoplex.

The final precision weight and its mapping to stiffness are

$$\begin{aligned} w_i(\phi) &= \text{clip}\Big(\frac{p_i(\phi)}{p_{\text{ref}}},\, 0,\, 1\Big) = \text{clip}\Big(\frac{1/(\sigma_i^2(\phi) + \epsilon_\sigma)}{p_{\text{ref}}},\, 0,\, 1\Big), \\ K_i^{\text{pos}}(\phi) &= K_{\text{floor}} + \big(K_{\max} - K_{\text{floor}}\big)\, w_i(\phi). \end{aligned} \tag{22}$$

$\sigma_i^2(\phi)$ is the cross-trial variance of the range-normalised signal at phase ϕ (dimensionless, so axes are comparable); ϵ_σ is a tiny constant avoiding division by zero; p_{ref} is a robust 95th-percentile normaliser *shared across all axes and channels*, so a true imprecise direction receives a true small weight $w_i \to 0$. The weight $w_i \in [0, 1]$ then scales the stiffness linearly between a floor K_{floor} [N m^{-1}] (kept non-zero so the arm can always execute the motion) and the maximum $K_{\max}$. Precision values at or above p_{ref} are clipped to $w_i = 1$. Using the 95th percentile rather than the absolute maximum prevents a single exceptionally low-variance sample from compressing all other weights towards zero. Figure 1 illustrates the result: the data-derived weights make the cut-depth axis comparatively stiff while lateral and force directions remain more compliant. The mapping is not parameter free: its percentile normalisation and stiffness bounds are fixed design choices.

The same procedure applied to the other two materials (Fig. 2) yields the *same qualitative structure* — depth regulated, force released — but with *material-specific magnitudes*, because the gains are computed from each material's own demonstration. This is the gain-side counterpart of the per-material force law of Section 6: one principle, three distinct parameter sets.

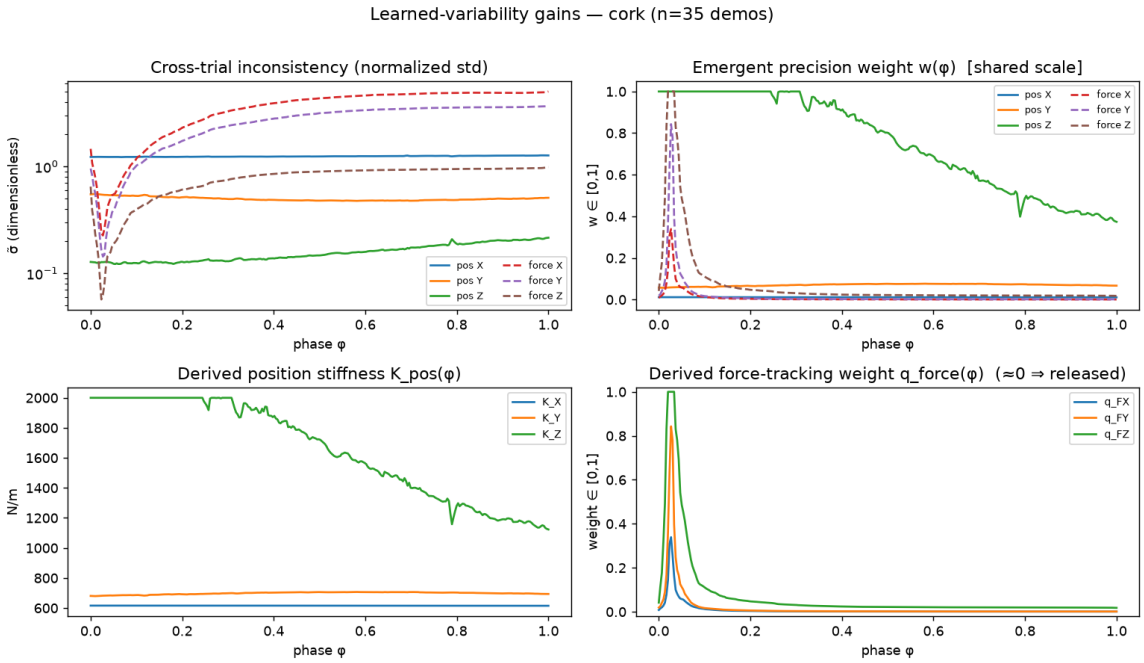


Figure 1: Gains learned from the cork demonstrations. Top-left: cross-trial inconsistency of each channel; top-right: the emergent precision weight $w(\phi)$; bottom: the resulting position stiffness $K^{\text{pos}}(\phi)$ and the (near-zero) force-tracking weight. The depth axis (blue) is regulated stiffly; force is released. *This split emerges from the data, not from tuning.*

3.3 The online quadratic program (QP) for the gains

At each control step, the gains are obtained from a small convex quadratic program (QP) that *blends* two objectives and respects two constraints:

$$\boxed{\begin{aligned} K^\star = \arg\min_{K\in\mathbb{R}^3} \; & q_f(\phi)\,\|K\odot\tilde{x} - F_d\|_2^2 \;+\; \big(1-q_f(\phi)\big)\,\|K - K^{\text{pos}}(\phi)\|_2^2 \\ \text{s.t.}\;\; & K_{\min} \le K \le K_{\max}, \qquad -F_i^{\text{bound}} \le K_i\,\tilde{x}_i \le F_i^{\text{bound}} \;\; \forall i. \end{aligned}} \tag{23}$$

Term by term:

- $\|K\odot\tilde{x} - F_d\|_2^2$ is the *force-consistency* cost ($\odot$ is element-wise product): it pulls the impedance force $K\odot\tilde{x}$ [N] to match the desired force F_d [N]. Its weight $q_f(\phi)\in[0,1]$ is the *force-precision* weight - small, because force is not reproducible (Sec. 2).
- $\|K - K^{\text{pos}}(\phi)\|_2^2$ pulls the stiffness towards the precision-derived value K^{pos} [$\text{N}\,\text{m}^{-1}$] of (22), weighted by $1-q_f$. Because $q_f\to 0$, this term dominates and the QP returns $K\approx K^{\text{pos}}$: the controller tracks the precision-weighted position skill.
- The box constraint $K_{\min}\le K\le K_{\max}$ keeps stiffness in the configured numerical range; hardware safety is not evaluated.
- The constraint $|K_i\tilde{x}_i|\le F_i^{\text{bound}}$ bounds the proportional residual-wrench component. The complete residual, including integral and damping terms, is clipped separately. Neither bound directly constrains measured contact force.

The damping follows from $K^\star$ by the fixed damping-ratio heuristic. The QP has three decision variables and is evaluated at every simulated control step; no wall-clock real-time guarantee is claimed. The raw solution is smoothed with the preceding gain and then reprojected onto the current error-dependent feasible interval before use.

Controlled comparison of learned variance structure. To test whether performance is associated with the selected precision allocation, the identical task and controller are evaluated under four gain assignments: (i) the identified precision; (ii) *inverted* precision (regulating directions that were inconsistent and relaxing those that were consistent); (iii) precision *permuted* across axes; and (iv) the original fixed-weight QP with learned gains disabled. The saved artifact historically calls the fourth condition "direct force", but it is not a measured-force feedback controller. The metric is the *cut-depth error* — the task-critical, demonstration-regulated axis — evaluated over 12 seeds with randomised contact parameters (Fig. 3, Table 3).

Let $w(\phi) = [w_1(\phi), w_2(\phi), w_3(\phi)]^\top \in [0,1]^3$ be the identified position-precision vector of (22), and define the affine normalisation of the Cartesian stiffness vector by

$$u(\phi) = \frac{K^{\text{pos}}(\phi) - K_{\min}\mathbf{1}}{K_{\max} - K_{\min}} \in [0,1]^3, \qquad K^{\text{pos}}(\phi) = K_{\min}\mathbf{1} + (K_{\max} - K_{\min})u(\phi). \tag{24}$$

Here $K^{\text{pos}} = [K_1^{\text{pos}}, K_2^{\text{pos}}, K_3^{\text{pos}}]^\top$ is the Cartesian stiffness vector, $\mathbf{1} \in \mathbb{R}^3$ is the all-ones vector, and the scalar denominator is the full admissible stiffness range. Thus $u_i = 0$ and $u_i = 1$ correspond respectively to $K_i^{\text{pos}} = K_{\min}$ and $K_i^{\text{pos}} = K_{\max}$; u is a dimensionless coordinate representation of the commanded stiffness rather than an additional controller state. Substitution of the learned gain law (22) into (24) gives

$$\begin{aligned} u_w(\phi) &= \eta_0\mathbf{1} + \eta_1 w(\phi), \\ \eta_0 &= \frac{K_{\text{floor}} - K_{\min}}{K_{\max} - K_{\min}}, \\ \eta_1 &= \frac{K_{\max} - K_{\text{floor}}}{K_{\max} - K_{\min}}. \end{aligned} \tag{25}$$

Accordingly, u_w denotes the normalized Cartesian stiffness induced by the precision weights obtained from the demonstrations.

The controlled comparison uses the exact transformations

$$\begin{aligned} u^{\text{true}}(\phi) &= u_w(\phi), \\ u^{\text{inv}}(\phi) &= \mathbf{1} - u_w(\phi), \qquad \Pi(\phi)^\top\Pi(\phi) = I, \\ u^{\text{perm}}(\phi) &= \Pi(\phi)u_w(\phi), \end{aligned} \tag{26}$$

The matched condition applies each learned component to its identified Cartesian axis. The inverted condition acts componentwise and satisfies

$$u_i^{\text{inv}}(\phi) = 1 - u_{w,i}(\phi), \qquad u_i^{\text{inv}} - u_j^{\text{inv}} = -\big(u_{w,i} - u_{w,j}\big), \tag{27}$$

thereby reversing the component ordering about the midpoint of the admissible normalized stiffness interval. The permuted condition preserves the instantaneous multiset of components of u_w but changes their Cartesian-axis assignment. These transformations distinguish sensitivity to the learned ordering from sensitivity to the learned axis association. The direct-force condition is not an allocation transformation in this space: it disables the learned-gain mode and applies the fixed force-QP weighting of the baseline controller.

Write the identified allocation as

$$u_w(\phi) = \begin{bmatrix} u_{w,x}(\phi) & u_{w,y}(\phi) & u_{w,z}(\phi) \end{bmatrix}^\top. \tag{28}$$

For a bijection $\pi_\phi : \{1,2,3\} \to \{1,2,3\}$, define

$$[\Pi(\phi)]_{jk} = \begin{cases} 1, & k = \pi_\phi(j), \\ 0, & \text{otherwise}, \end{cases} \qquad j,k \in \{1,2,3\}. \tag{29}$$

It follows that

$$[\Pi(\phi)u_w(\phi)]_j = u_{w,\pi_\phi(j)}(\phi), \qquad \Pi(\phi)^\top\Pi(\phi) = I, \qquad \Pi(\phi)^{-1} = \Pi(\phi)^\top. \tag{30}$$

Consequently, $\Pi(\phi)$ preserves the Euclidean norm and the multiset of components of $u_w(\phi)$ while changing their Cartesian-axis assignment. The dependence on ϕ permits the assignment to vary over task phase. In the implemented shuffled condition, ϕ determines the pseudorandom seed used to generate π_ϕ. This comparison isolates the effect of the learned axis assignment while preserving the instantaneous set of stiffness allocations.

Proposition 2 (Uniqueness of the matched allocation under the precision-deviation functional)**.** *Consider the precision-allocation functional*

$$\mathcal{J}[u] = \frac{1}{2}\int_0^1 \big(u(\phi) - u_w(\phi)\big)^\top H(\phi)\big(u(\phi) - u_w(\phi)\big)\, d\phi, \qquad H(\phi) = H(\phi)^\top \succ 0. \tag{31}$$

Among all measurable allocations $u : [0,1] \to [0,1]^3$, the matched assignment $u^{\text{true}} = u_w$ is the unique minimiser of (31)*, up to sets of zero measure. The excess costs of the inverted and permuted structures are*

$$\mathcal{G}_{\text{inv}} = \frac{1}{2}\int_0^1 (\mathbf{1} - 2u_w)^\top H(\mathbf{1} - 2u_w)\, d\phi, \tag{32}$$

$$\mathcal{G}_{\text{perm}} = \frac{1}{2}\int_0^1 (\Pi u_w - u_w)^\top H(\Pi u_w - u_w)\, d\phi. \tag{33}$$

Both are strictly positive whenever the transformation changes a task-weighted component of u_w on a phase interval of non-zero measure.

Proof. Since $H(\phi) \succ 0$, its smallest eigenvalue $h_{\min}(\phi) = \lambda_{\min}(H(\phi))$ is positive and

$$\mathcal{J}[u] \geq \frac{1}{2}\int_0^1 h_{\min}(\phi)\, \|u(\phi) - u_w(\phi)\|_2^2\, d\phi \geq 0. \tag{34}$$

The lower bound is attained by $u = u_w$, for which $\mathcal{J}[u_w] = 0$. Equality for another allocation requires $u(\phi) = u_w(\phi)$ almost everywhere, which proves uniqueness. Substitution of $u^{\text{inv}} = \mathbf{1} - u_w$ and $u^{\text{perm}} = \Pi u_w$ into (31) gives (32) and (33). Positive definiteness of H makes either integral strictly positive when its difference vector is non-zero on a set of non-zero measure. □

Define the allocation error

$$e_u(\phi) = u(\phi) - u_w(\phi) = \begin{bmatrix} e_x(\phi) & e_y(\phi) & e_z(\phi) \end{bmatrix}^\top. \tag{35}$$

Then the integrand of (31) is the quadratic form

$$\frac{1}{2}e_u^\top H e_u = \frac{1}{2}\sum_{j=1}^{3}\sum_{k=1}^{3} H_{jk}(\phi)e_j(\phi)e_k(\phi). \tag{36}$$

Here $H : [0,1] \to \mathbb{S}^3_{++}$ is a measurable, phase-dependent task-relevance metric on the allocation space. Its diagonal entries weight axis-specific allocation errors, whereas its off-diagonal entries encode cross-axis coupling. The condition $H(\phi) \succ 0$ means

$$v^\top H(\phi)v > 0 \quad \text{for every non-zero } v \in \mathbb{R}^3, \tag{37}$$

and is the condition required for the uniqueness result in Proposition 2. For the cutting task, greater depth-axis relevance may be represented through the corresponding entries of $H(\phi)$.

The matrix $H(\phi)$ is an assumed metric defining the quadratic allocation functional; it is not estimated from the demonstrations, numerically specified in the experiments, or evaluated by the controller. Consequently, Proposition 2 establishes only that u_w uniquely minimizes the defined weighted deviation from u_w. It does not establish optimality with respect to cutting error, interaction force, or a general closed-loop performance functional. The physical consequence of the matched allocation is instead evaluated by the paired closed-loop experiment below.

For controller condition c and randomised-physics seed s, the reported cut-depth metric is

$$E^z_{s,c} = 10^3 \sqrt{\frac{1}{t_{f,s}} \int_0^{t_{f,s}} \left(x^{(s,c)}_{d,z}(t) - x^{(s,c)}_z(t)\right)^2 dt} \quad [\text{mm}], \tag{38}$$

where $t_{f,s}$ is the episode duration, $x_{d,z}$ is desired scalpel depth, x_z is measured depth, and the factor 10^3 converts metres to millimetres. The same randomised mass, friction, contact parameters and initial perturbation are used for all conditions within seed s, so the paired difference

$$G_{s,c} = E^z_{s,c} - E^z_{s,\text{true}} \tag{39}$$

removes seed-specific physics shared by the two runs. Positive $G_{s,c}$ means that condition c has larger cut-depth error than the identified precision assignment.

Gain assignment	depth RMSE (mm)	identified reduction	Wilcoxon p
Identified precision	$\mathbf{29.39 \pm 46.72}$	—	—
Permuted precision	31.25 ± 47.54	6.0%	0.001
Inverted precision	44.39 ± 47.65	33.8%	0.001
Direct force-tracking	47.84 ± 51.73	38.6%	0.002

Table 3: The identified precision has lower mean cut-depth error than all three comparators. It is lower in all 11 paired seeds against the permuted and inverted assignments ($p = 0.001$), and in 10 of 11 against the direct-force baseline ($p = 0.002$). The large dispersion requires interpreting this as a paired allocation result rather than a claim of uniformly small absolute error.

For the $n_{\text{pair}} = 11$ seeds that were stable in every paired condition, the table reports

$$\bar{E}_c = \frac{1}{n_{\text{pair}}} \sum_{s=1}^{n_{\text{pair}}} E^z_{s,c}, \qquad s_c = \sqrt{\frac{1}{n_{\text{pair}} - 1} \sum_{s=1}^{n_{\text{pair}}} \left(E^z_{s,c} - \bar{E}_c\right)^2}, \tag{40}$$

where $\bar{E}_c$ and s_c are respectively the sample mean and sample standard deviation for condition c. The percentage shown in the third column is the error reduction achieved by the identified assignment relative to comparator c,

$$R_c = 100\, \frac{\bar{E}_c - \bar{E}_{\text{true}}}{\bar{E}_c} \quad [\%], \tag{41}$$

which gives 6.0%, 33.8% and 38.6% for the permuted, inverted and direct-force conditions. This denominator explains why these percentages are reductions relative to each comparator rather than percentage increases relative to the identified mean.

Proposition 3 (Paired dominance in the evaluated seeds)**.** *For the permuted and inverted comparator conditions, the evaluated pairs satisfy $G_{s,c} > 0$ for all $s \in \{1, \ldots, 11\}$. The exact two-sided paired Wilcoxon signed-rank test therefore gives*

$$p_{\text{exact}} = 2^{1-n_{\text{pair}}} = 2^{-10} = 0.0009765625 \approx 0.001. \tag{42}$$

Proof. Let ϱ_s be the rank of $|G_{s,c}|$ among the n_{pair} non-zero paired differences and define the negative signed-rank statistic

$$W_c^- = \sum_{s=1}^{n_{\text{pair}}} \varrho_s \mathbb{1}_{\{G_{s,c}<0\}}. \tag{43}$$

For these two comparators every observed difference is positive, so $W_c^- = 0$. Under the Wilcoxon null hypothesis of a distribution symmetric about zero, each of the $2^{n_{\text{pair}}}$ sign assignments is equally likely. Only the all-positive and all-negative assignments are at least as extreme as the observed one, so the two-sided exact probability is $2/2^{n_{\text{pair}}} = 2^{1-n_{\text{pair}}}$. Substitution of $n_{\text{pair}} = 11$ yields (42). The direct-force baseline is excluded from this proposition because one of its 11 paired differences is negative; its separately computed exact test gives $p = 0.002$. □

Proposition 3 is an exact statement about the evaluated paired seeds and rejects the zero-median-difference null at the reported sample size. Together with Proposition 2, it links the analytical consequence of matching the learned precision allocation to the observed closed-loop reduction in the task-critical error, without treating the finite simulation sample as a universal proof for all contact conditions.

4 State-driven task parametrization

4.1 Theoretical motivation

Contact-rich skill execution involves two distinct objects: the geometric and interaction structure of the skill, and the rate at which that structure is traversed. A purely temporal reference conflates them by imposing $x_d(t) = x_r(t/T_d)$. Under free motion this may be adequate; under contact, however, the realised state is affected by material deformation, friction, actuator limits, external perturbations and passivity intervention. The temporal phase can therefore continue to advance while the physical system remains near an earlier task state.

Let

$$z(t) = \begin{bmatrix} x(t)^\top & \|F(t)\|_2 \end{bmatrix}^\top, \qquad z_r(\phi) = \begin{bmatrix} x_r(\phi)^\top & \|F_r(\phi)\|_2 \end{bmatrix}^\top \tag{44}$$

denote the measured and reference task states, respectively. The phase consistent with the realised task state is the local projection

$$\phi^\star(t) \in \arg\min_{\varphi \in \mathcal{I}(t)} \left\| W\big(z_r(\varphi) - z(t)\big) \right\|, \tag{45}$$

where W is a dimensional normalisation/weighting matrix and $\mathcal{I}(t) \subseteq [0,1]$ restricts the admissible branch of the reference curve. For a clock phase $\phi_c(t) = t/T_d$, define the synchronisation error $e_\phi(t) = \phi_c(t) - \phi^\star(t)$. If x_r is differentiable, the reference displacement induced solely by this loss of synchronisation satisfies the first-order relation

$$x_r(\phi_c) - x_r(\phi^\star) = x_r'(\phi^\star) e_\phi + \mathcal{O}(e_\phi^2). \tag{46}$$

Thus a phase error is converted directly into a Cartesian reference error, scaled by the local path tangent. During contact, this error is further mapped through the impedance into a commanded wrench. Continuing a clock-driven reference during arrested motion can consequently increase both tracking error and demanded interaction power, even when the geometric prior itself is appropriate.

State-driven parametrisation addresses this synchronisation problem by treating phase as an internal progress coordinate inferred from the realised task state, rather than as elapsed time alone. This has three theoretical consequences. First, it separates the phase-indexed task manifold $\mathcal{M} = \{z_r(\phi) : \phi \in [0,1]\}$ from its temporal traversal law. Second, local projection onto $\mathcal{M}$ makes reference selection dependent on physical task progress. Third, constraining the phase law to be non-negative preserves the causal ordering of approach, engagement, cutting and termination.

For the present controller, state projection, nominal temporal progression and the energy-tank gate are combined in a sampled-data phase law; the precise construction follows.

4.2 Phase-domain skill representation

Let the learned behaviour prior be the phase-indexed curve

$$\mathcal{B}(\phi) = \big(x_r(\phi),\, x_r'(\phi),\, F_r(\phi),\, K(\phi),\, D(\phi)\big), \qquad \phi \in [0,1], \tag{47}$$

where a prime denotes differentiation with respect to phase.

4.2.1 Root correction of demonstration registration

The demonstrations in this study encode a nominally straight cut. Direct use of the GMR position mean introduced a small transverse curvature: for cork, the 300-sample mean has a 7.066 mm RMS orthogonal distance from its three-dimensional principal line (maximum 13.010 mm), although its path length is only 2.25% greater than its endpoint distance. Trimming symmetric fractions from the record did not reveal a distinct approach or withdrawal segment; the curvature persists over the central cutting samples. The force magnitude is already 42.586 N at the first sample, remains non-zero throughout, and reaches the cutting plateau toward the end. Hence there is no force-based evidence that the stored prior contains an initial free-space approach. The curvature is therefore not treated as part of the intended task geometry.

The corrected learning pipeline does not truncate demonstrations by a shared sample index. For demonstration i, it first computes endpoint progress

$$s_{i,k} = \mathrm{clip}_{[0,1]} \frac{(p_{i,k} - p_{i,0})^\top (p_{i,n_i-1} - p_{i,0})}{\|p_{i,n_i-1} - p_{i,0}\|_2^2}, \qquad \bar{s}_{i,k} = \max_{\ell \le k} s_{i,\ell}, \tag{48}$$

and interpolates every pose, wrench and variability channel on a common uniform grid in $\bar{s}_i$. Thus equal phase denotes equal spatial progress rather than equal sample number.

A single structured GMM/GMR is learned from all materials. Material identity is not represented by the ordinal scalar $1, 2, 3$. Instead, with cork as reference, the categorical contrast is

$$c(\text{cork}) = (0,0), \quad c(\text{penoplex}) = (1,0), \quad c(\text{PVC}) = (0,1),$$

and the conditioning vector is

$$u(\phi, m) = \begin{bmatrix} \phi & c_1(m) & c_2(m) & \phi c_1(m) & \phi c_2(m) \end{bmatrix}^\top. \tag{49}$$

The interaction terms permit material-dependent affine endpoints and cutting directions without imposing an artificial ordinal distance between materials. For fixed m, the products $\phi c_j(m)$ remain affine functions of phase, but allow their coefficients to differ between materials.

4.2.2 Shared structured GMM/GMR

Let b collect orientation, wrench and twist, and define the two random vectors

$$v^x = \begin{bmatrix} u^\top & x^\top \end{bmatrix}^\top, \qquad v^b = \begin{bmatrix} u^\top & b^\top \end{bmatrix}^\top. \tag{50}$$

Both blocks use all $N_{\text{demo}} = 35 + 26 + 35 = 96$ demonstrations. They are parts of one structured material-conditioned prior; no block is fitted separately for cork, penoplex or PVC. Before estimation, each scalar training variable is standardised using statistics of the complete multi-material training set:

$$\tilde{v}_j = \frac{v_j - \mu_j^{\text{data}}}{\sigma_j^{\text{data}}}. \tag{51}$$

This prevents variables measured in metres, degrees, newtons and newton-metres from dominating the likelihood solely through their units.

For block $r \in \{x, b\}$, the joint density is

$$p_r(\tilde{u}, \tilde{y}_r) = \sum_{q=1}^{K_r} \pi_{r,q} \mathcal{N}\left(\begin{bmatrix} \tilde{u} \\ \tilde{y}_r \end{bmatrix} ; \begin{bmatrix} \mu_{r,q}^u \\ \mu_{r,q}^y \end{bmatrix}, \begin{bmatrix} \Sigma_{r,q}^{uu} & \Sigma_{r,q}^{uy} \\ \Sigma_{r,q}^{yu} & \Sigma_{r,q}^{yy} \end{bmatrix} \right), \tag{52}$$

where $y_x = x$ and $y_b = b$. Full-covariance mixtures are estimated by expectation–maximisation with covariance regularisation $10^{-5} I$, three initialisations, random seed 42 and at most 1000 EM iterations. The positional block uses $K_x = 1$; the nonlinear interaction block uses $K_b = 15$.

Runtime priors are generated by Gaussian mixture regression, not by component classification. For a requested $\tilde{u}$, component q gives

$$\mu_{r,q}^{y|u}(\tilde{u}) = \mu_{r,q}^y + \Sigma_{r,q}^{yu} (\Sigma_{r,q}^{uu})^{-1} (\tilde{u} - \mu_{r,q}^u), \tag{53}$$

$$h_{r,q}(\tilde{u}) = \frac{\pi_{r,q} \mathcal{N}(\tilde{u}; \mu_{r,q}^u, \Sigma_{r,q}^{uu})}{\sum_{j=1}^{K_r} \pi_{r,j} \mathcal{N}(\tilde{u}; \mu_{r,j}^u, \Sigma_{r,j}^{uu})}, \tag{54}$$

$$\hat{\tilde{y}}_r(\tilde{u}) = \sum_{q=1}^{K_r} h_{r,q}(\tilde{u}) \mu_{r,q}^{y|u}(\tilde{u}). \tag{55}$$

The final prediction is returned to physical units by inverting (51). Equation (55) is implemented directly because the external GMR package used by the legacy notebook is incompatible with the installed numerical-library version; the equation is the standard analytical GMR conditional mean.

For $K_x = 1$, $h_{x,1} = 1$. Since $u(\phi, m)$ is affine in ϕ for fixed material, (55) makes $\hat{x}(\phi, m)$ affine in ϕ. Straightness is therefore a property of the trained conditional position model, rather than a controller-side correction. The 15-component block retains the capacity to represent nonlinear force, orientation and twist profiles.

Position prior	orthogonal RMS [mm]	maximum [mm]	length/chord
Legacy cork GMR	7.066	13.010	1.022524
Legacy penoplex GMR	0.762	2.122	1.000204
Legacy PVC GMR	0.405	1.270	1.000099
Corrected shared GMR, each material	$< 10^{-6}$	$< 10^{-6}$	1.000000

Table 4: Straightness audit of the saved conditional position priors. Distance is measured orthogonally to the three-dimensional principal line. The corrected values are evaluated before controller loading or workspace mapping.

The conditioned priors are exported per material only as runtime data arrays; these files are evaluations of the same shared model, not independently fitted models. The controller consumes them directly and contains no downstream straightening operation. The demonstrated result concerns categorical conditioning on the three materials included in training; interpolation or zero-shot prediction for an unseen material is not claimed.

For any absolutely continuous, non-decreasing phase $\phi(t)$, the commanded position and its physically consistent velocity satisfy

$$x_d(t) = x_r(\phi(t)), \qquad \dot{x}_d(t) = x_r'(\phi(t)) \dot{\phi}(t). \tag{56}$$

Temporal scaling changes $\dot{x}_d$ without changing the image of the path. If the demonstration has duration T_d and the requested dimensionless speed multiplier is $s_\phi > 0$, the ungated nominal phase rate is

$$\dot{\phi}_{\text{nom}} = \frac{s_\phi}{T_d}, \qquad T_{\text{exec}}^{\text{nom}} = \frac{T_d}{s_\phi}. \tag{57}$$

This defines the nominal traversal law prior to state and passivity adaptation.

4.3 State-to-phase observation

At controller sample k, let ϕ_k be the phase used to evaluate the prior, x_k the measured TCP position, and F_k the filtered measured interaction force. The stored reference contains N uniformly spaced knots

$$\Phi_N = \left\{ \phi^{(j)} = \frac{j}{N-1} : j = 0, \dots, N-1 \right\}. \tag{58}$$

Rather than searching the entire curve, the implementation forms the local candidate set

$$\mathcal{C}_k = \left\{ \phi^{(j)} \in \Phi_N : \max(0, \phi_k - \ell_\phi^-) \le \phi^{(j)} \le \min(1, \phi_k + \ell_\phi^+) \right\}, \tag{59}$$

where $\ell_\phi^- = \ell_\phi^+ = 0.1$. The backward branch permits measured lag to retard the reference; monotonicity is imposed by the phase dynamics rather than by excluding earlier candidates. The normalising scales used in the implementation are

$$\rho_x = \|x_r(1) - x_r(0)\|_2 + \varepsilon_x, \qquad \rho_F = \max_{\phi^{(j)} \in \Phi_N} \left\| F_r(\phi^{(j)}) \right\|_2 + \varepsilon_F, \tag{60}$$

with $\varepsilon_x, \varepsilon_F > 0$. Define the contact indicator

$$c_k = \mathbb{1}_{\{\|F_k\|_2 > F_{\text{contact}}\}}. \tag{61}$$

The phase-observation cost is

$$J_k(\varphi) = \begin{cases} \dfrac{\|x_r(\varphi) - x_k\|_2}{\rho_x}, & c_k = 0, \\ \dfrac{1}{2} \dfrac{\|x_r(\varphi) - x_k\|_2}{\rho_x} + \dfrac{1}{2} \dfrac{|\|F_r(\varphi)\|_2 - \|F_k\|_2|}{\rho_F}, & c_k = 1. \end{cases} \tag{62}$$

The observed progress is the discrete nearest point

$$\hat{\phi}_k \in \arg\min_{\varphi \in \mathcal{C}_k} J_k(\varphi), \tag{63}$$

with the first minimising knot resolving ties. Retaining the position term during contact mitigates non-uniqueness of the force-magnitude map $\phi \mapsto \|F_r(\phi)\|_2$. The bounded search interval selects a local branch when the reference curve is not globally injective.

4.4 Sampled-data phase dynamics

Let h_c denote the controller period and let the energy-tank gate $\gamma_{T,k} \in [0, 1]$ be defined by (75). The nominal increment available at sample k is

$$\Delta\phi_k^{\text{nom}} = h_c \gamma_{T,k} \dot{\phi}_{\text{nom}} = \gamma_{T,k} \frac{s_\phi h_c}{T_d}. \tag{64}$$

The raw state correction and its implemented increment limiter are

$$r_k = h_c k_\phi (\hat{\phi}_k - \phi_k), \qquad k_\phi = 2.0\,[\text{s}^{-1}], \tag{65}$$

$$\delta\phi_k = \text{sat}_{[-\Delta\phi_k^{\text{nom}},\, 2\Delta\phi_k^{\text{nom}}]}(r_k), \tag{66}$$

where $\text{sat}_{[a,b]}(z) = \min\{b, \max\{a, z\}\}$. The complete recursion is therefore

$$\boxed{\phi_{k+1} = \Pi_{[0,1]}(\phi_k + \Delta\phi_k^{\text{nom}} + \delta\phi_k)\,,} \tag{67}$$

where $\Pi_{[0,1]}(z) = \min\{1, \max\{0, z\}\}$. All terms in the recursion are dimensionless phase increments.

Proposition 4 (Monotonicity and bounded phase rate)**.** *For* $\phi_0 \in [0,1]$*,* $\gamma_{T,k} \in [0,1]$*, and* $\dot{\phi}_{\mathrm{nom}} \geq 0$*, recursion* (67) *satisfies*

$$0 \leq \phi_k \leq 1, \qquad 0 \leq \phi_{k+1} - \phi_k \leq 3h_c\gamma_{T,k}\dot{\phi}_{\mathrm{nom}}. \tag{68}$$

Proof. Projection gives $0 \leq \phi_{k+1} \leq 1$. From (66), $-\Delta\phi_k^{\mathrm{nom}} \leq \delta\phi_k \leq 2\Delta\phi_k^{\mathrm{nom}}$, hence the unprojected increment belongs to $[0, 3\Delta\phi_k^{\mathrm{nom}}]$. Projection at the upper endpoint can only reduce a non-negative increment, proving (68). □

Setting $k_\phi = 0$ removes state matching and gives a tank-gated clock. Setting additionally $\gamma_{T,k} = 1$ gives uniform time parametrisation. For $k_\phi > 0$, the bidirectional state observation retards phase when $\hat{\phi}_k < \phi_k$ and accelerates it when $\hat{\phi}_k > \phi_k$. The lower saturation limit permits complete cancellation of the nominal increment but precludes reversal; the upper limit bounds the total rate by three times the tank-gated nominal rate. In addition, as $T_k \downarrow T_{\min}$, $\gamma_{T,k} \downarrow 0$, which drives both $\Delta\phi_k^{\mathrm{nom}}$ and its permitted correction to zero. This separation prevents phase reversal while coupling task timing to the available energy budget. The implemented construction is therefore a *bidirectionally corrected, tank-gated state-dependent parametrisation.* It provides temporal scaling through s_ϕ, measured-state correction through (63), monotone bounded progression through (67), and energy-dependent slowing through $\gamma_{T,k}$.

4.5 Controlled phase–state desynchronisation

Workpiece motion changes contact geometry but does not necessarily create a repeatable lag along the cutting path. A separate exogenous perturbation was therefore introduced to test the causal property for which state-driven parametrisation was designed. Let

$$\hat{t}_r(\phi) = \frac{x_r'(\phi)}{\|x_r'(\phi)\|_2 + \varepsilon_t} \tag{69}$$

denote the regularised unit tangent to the reference path. During the prescribed interval $[t_a, t_b]$, the MuJoCo external-force input on the scalpel body is augmented by

$$f_h(t,\phi) = -F_h\,\hat{t}_r(\phi)\,\mathbb{1}_{\{t_a \leq t \leq t_b\}}, \qquad F_h = 120\,[\mathrm{N}], \quad (t_a, t_b) = (0.9, 1.4)\,[\mathrm{s}]. \tag{70}$$

This *holdback force* acts opposite to instantaneous path progression and is added to, rather than substituted for, the identified material reaction. It enters the robot dynamics through the physical external-wrench channel; it does not modify ϕ, $\hat{\phi}$ or $\dot{\phi}$ numerically. Its causal route is therefore

$$f_h \longrightarrow (x, \dot{x}, F) \longrightarrow \hat{\phi} \longrightarrow \dot{\phi}.$$

The perturbation is not part of nominal cutting or of the learned controller. It is a validation input analogous to an impulse or step disturbance in a closed-loop robustness experiment. Runtime configuration consequently defaults it to disabled; its enable flag, amplitude and time interval are independent of the switch selecting state-driven or time-driven phase.

4.6 Factorial runtime ablation

The evaluation crossed three binary factors: static versus vertically moving workpiece, nominal versus holdback-perturbed execution, and state-driven versus time-driven phase. The time-driven condition sets the state-correction term to zero while retaining the learned variable impedance, material model, energy-tank gate and torque projection. Thus the comparison does not remove a safety mechanism. Five paired randomised-physics seeds were used; within each scenario and seed the two phase laws received identical friction, contact solver parameters, material mass and initial joint perturbation.

The static and moving conditions use the same world-frame desired position and velocity trajectories. Workpiece motion is applied only to the simulated material joint and is not added to either reference; it is therefore an external geometric perturbation, matching the experimental convention used for the holdback force. The material surface used by the contact model remains live, so only the environment geometry changes.

Table 5: Factorial task-parametrisation comparison. Entries are mean ± standard deviation over $N = 5$ paired seeds; S and T denote state-driven and time-driven phase, respectively.

Workpiece	Holdback	Cut-depth RMSE [mm]		Position RMSE [mm]	
		S	T	S	T
Static	off	2.485 ± 1.029	2.538 ± 1.075	12.250 ± 10.461	12.533 ± 11.229
Static	on	3.165 ± 0.795	3.694 ± 1.153	36.422 ± 21.713	48.449 ± 42.406
Moving	off	9.557 ± 8.805	9.631 ± 8.981	21.602 ± 19.391	22.573 ± 22.839
Moving	on	10.821 ± 7.495	12.209 ± 7.085	73.324 ± 15.578	96.417 ± 28.339

All 40 executions completed and the torque-port power-violation fraction was zero. The state-driven law has lower mean cut-depth RMSE in three cells but higher error in moving nominal execution. Full Cartesian RMSE is lower in the two perturbed cells and higher in both nominal cells. None of the paired cut-depth comparisons is significant ($p \in [0.0625, 0.8125]$). The large seed-to-seed dispersion, including a moving workpiece outlier, precludes a general accuracy-improvement claim. The supported result is narrower: state feedback changes phase progression while preserving the implemented torque-port constraint and task completion. It does not reject a 120 N holdback with small Cartesian error, nor does it establish invariance to workpiece motion. Because learned precision-dependent impedance, the per-material cutting model, force saturation, energy tank and torque projection remain active in every cell, completion and constraint satisfaction evaluate the integrated proposed architecture in both workpiece conditions. The within-cell S–T difference, by contrast, isolates only the incremental effect of state-dependent timing; it must not be attributed to the other contributions.

5 Conditional stability analysis and discrete torque-port energy accounting

The scalpel obeys the task-space dynamics $\Lambda\ddot{x} + C\dot{x} + g = f + f_{\text{ext}}$, where Λ [kg] is the Cartesian inertia and f_{ext} [N] the environment force. Substituting the control law (13) (with gravity/Coriolis cancelled by (16)) gives the *error dynamics*

$$\Lambda\ddot{\tilde{x}} + \big(C + D(\phi)\big)\dot{\tilde{x}} + K(\phi)\,\tilde{x} = f_{\text{ext}}, \tag{71}$$

a (time-varying) mass–spring–damper in the tracking error driven by the contact force.

5.1 Ideal continuous-time passivity condition

Proposition 5 (Continuous-time tank passivity under exact accounting)**.** *At the Cartesian contact port, the input–output map $f_{ext} \mapsto \dot{x}$ is passive under the following tank dynamics: With the storage function (an energy, in joules) $\mathcal{S} = \frac{1}{2}\dot{\tilde{x}}^\top \Lambda \dot{\tilde{x}} + \frac{1}{2}\tilde{x}^\top K \tilde{x} + T$ and a non-negative* energy tank $T \geq T_{\min}$ *whose dynamics collect the power dissipated by damping and compensate the power injected by the time-varying gains and moving reference,*

$$\dot{T} = \underbrace{\dot{\tilde{x}}^\top D\,\dot{\tilde{x}}}_{\textit{collected}\ [\text{W}]} - \underbrace{P_{inj}}_{[\text{W}]}, \qquad P_{inj} = \tfrac{1}{2}\tilde{x}^\top \dot{K}\,\tilde{x} + \tilde{x}^\top K\,\dot{x}_d, \tag{72}$$

the loop satisfies $\dot{\mathcal{S}} \leq f_{ext}^\top \dot{x}$ as long as $T > T_{\min}$.

Interpretation: $\mathcal{S}$ is the total stored energy (kinetic + elastic + tank). The two indefinite, potentially energy-injecting terms — the rate of change of stored elastic energy from a rising stiffness ($\frac{1}{2}\tilde{x}^\top \dot{K}\tilde{x}$) and the power supplied to a moving reference ($\tilde{x}^\top K\dot{x}_d$), both in watts — are drawn from the tank, which is recharged by the power dissipated through damping. When the tank reaches its lower bound, stiffness growth is inhibited ($\dot{K} \leq 0$) and the phase is frozen ($\dot{\phi} \to 0$), so the controller cannot generate energy, instead, it reduces the motion velocity to avoid instant change of dynamics. This is an ideal continuous-time design condition. The implemented sampled-data torque-port balance is stated separately below.

5.2 Implemented discrete torque-port tank

The simulation controller enforces the tank constraint at the joint torque port, where the mechanical power supplied by the actuators is $P_\tau = \dot{q}^\top \tau$. At every control step the nominal torque

$$\tau_{\text{nom}} = \tau_{\text{task}} + \tau_{\text{null}} + \tau_{\text{bias}}, \qquad \tau_{\text{task}} = J^\top (JJ^\top + \lambda^2 I)^{-1} f, \qquad \tau_{\text{null}} = N\tau_{\text{posture}}, \qquad \tau_{\text{bias}} = g + C\dot{q}$$

is assembled from the task, null-space and model-compensation components. The task torque $\tau_{\text{task}}\,[\text{N m}]$ maps the Cartesian impedance command $f\,[\text{N}]$ to the joints through the damped Jacobian inverse, where $J \in \mathbb{R}^{3\times n}$ satisfies $\dot{x} = J\dot{q}$, λ^2 is the damped-least-squares regulariser, I is the identity matrix and n is the number of joints. The posture command $\tau_{\text{posture}}\,[\text{N m}]$ attracts the arm towards its home configuration and damps joint motion. Its projection $\tau_{\text{null}} = N\tau_{\text{posture}}$, with $N = I - J^\top (JJ^\top + \lambda^2 I)^{-1} J$, acts in the redundant null space and therefore satisfies $JN \approx 0$ without materially disturbing the Cartesian cutting task. The bias term $\tau_{\text{bias}}\,[\text{N m}]$ contains the gravity torque g and the Coriolis/centrifugal torque $C\dot{q}$. Their sum $\tau_{\text{nom}}\,[\text{N m}]$ is the nominal joint-torque command prior to passivity enforcement.

The nominal command is projected onto the half-space of admissible torques:

$$\boxed{\tau_{\text{safe}} = \arg\min_{\tau} \|\tau - \tau_{\text{nom}}\|_2^2 \quad \text{s.t.} \quad \dot{q}^\top \tau \leq P_{\max}, \qquad P_{\max} = \frac{T - T_{\min}}{h_c}.} \tag{73}$$

Here $\tau_{\text{safe}}\,[\text{N m}]$ is the projected actuator command, $q\,[\text{rad}]$ and $\dot{q}\,[\text{rad s}^{-1}]$ are the joint position and velocity, and $\dot{q}^\top \tau\,[\text{W}]$ is the mechanical power supplied at the joint port. The scalar $P_{\max}\,[\text{W}]$ is the power available from the usable tank energy $T - T_{\min}$ over one control period $h_c\,[\text{s}]$. Consequently, the constraint bounds positive energy delivery by the controller, while negative joint power corresponds to energy absorption. The quadratic objective retains the nominal control action as closely as possible in the Euclidean torque norm. If $\dot{q}^\top \tau_{\text{nom}} \leq P_{\max}$, then $\tau_{\text{safe}} = \tau_{\text{nom}}$. Otherwise the Euclidean projection has the closed form

$$\tau_{\text{safe}} = \tau_{\text{nom}} - \frac{\dot{q}^\top \tau_{\text{nom}} - P_{\max}}{\dot{q}^\top \dot{q} + \epsilon_\tau}\, \dot{q}, \tag{74}$$

where $\epsilon_\tau > 0$ is a small numerical regulariser. The correction lies along $\dot{q}$, the direction that changes instantaneous joint power, and is therefore the minimum-norm torque modification that satisfies the power bound. This expression is also used as a fallback if the numerical QP does not return a solution.

The power limit is computed from T_k, the torque is projected, and storage is then settled from the delivered joint power:

$$P_k = \dot{q}_k^\top \tau_{\text{safe},k}, \qquad T_{k+1} = \text{clip}(T_k - h_c P_k,\, T_{\min}, T_{\max}), \qquad \gamma_{T,k+1} = \tanh\left(\frac{T_{k+1} - T_{\min}}{T_{\text{ref}}}\right), \tag{75}$$

where $T_k\,[\text{J}]$ and $T_{k+1}\,[\text{J}]$ are the tank energies at consecutive samples, $T_{\min}\,[\text{J}]$ is the protected lower bound, $T_{\max}\,[\text{J}]$ is the storage ceiling, and $\text{clip}(\cdot)$ enforces these bounds. Positive commanded joint power depletes storage; negative power recharges it, with excess absorbed energy discarded

at $T_{\max}$. The projection ensures $h_c P_k \le T_k - T_{\min}$ for positive P_k. The implementation still logs damping, gain-variation and reference-motion powers as diagnostics, but does not use them for a second tank update. The reference energy T_{ref} [J] sets the transition scale of the phase gate: $\gamma_{T,k+1} \in [0,1]$ approaches zero as the usable tank energy $T_{k+1} - T_{\min}$ vanishes and modulates the phase rate according to $\dot{\phi} = \gamma_T \dot{\phi}_{\mathrm{nom}}$. Increasing T_{ref} makes this transition more gradual over the available energy range.

The resulting closed-loop sequence couples the timing, impedance and joint-torque layers. At interval k, T_k determines $\gamma_{T,k}$ and the planned increment (67); the current phase selects x_d, $\dot{x}_d$, $K(\phi)$ and $D(\phi)$, from which the impedance law produces f. The current tank determines $P_{\max}$, and the task mapping, null-space term, dynamic compensation and projection produce τ_{safe} satisfying $\dot{q}^\top \tau_{\mathrm{safe}} \le P_{\max}$. The delivered power then produces T_{k+1} and the next phase gate through (75). The nominal experiments use $T_0 = 50\,\mathrm{J}$, $T_{\max} = 100\,\mathrm{J}$ and $T_{\min} = 0.001\,\mathrm{J}$.

Controller	completion	min. tank (J)	budget violation	max margin (W)
No projection	100%	0.001	69.8%	+24,586
Tank projection	100%	0.001	0.0%	0
Tank + phase gate	0% ($\phi_f = 0.024$)	0.001	0.0%	0

Table 6: Torque-port storage and power validation under positive-power injection. Projection enforces the sampled bound. With phase gating and the stress-test's 1 J tank, depleted storage halts the task rather than violating the bound.

This is an empirical validation of the implemented sampled storage and power accounting, not a universal passivity or completion proof. The reconstructed Cartesian residual $\dot{V} - f_{\mathrm{ext}}^\top \dot{x}$ is still plotted as a diagnostic in Fig. 9, but it depends on finite differences, force-estimator sign conventions and an approximate Cartesian inertia; the implemented guarantee is therefore stated and validated at the torque port $\dot{q}^\top \tau$.

5.3 Sufficient exponential-stability conditions for idealised error dynamics

The stability argument is first established for nominal regulation and is then extended to phase-varying gains and bounded perturbations.

Assumption 1 (Nominal regulation conditions)**.** *The compensated Cartesian inertia satisfies $\Lambda = \Lambda^\top \succ 0$ and is constant over the local analysis interval; the frozen gains satisfy $K = K^\top \succ 0$ and $D = D^\top \succ 0$; gravity and Coriolis terms are compensated; and the nominal result is evaluated with zero external input and zero unmodelled reference acceleration.*

Under Assumption 1, defining $e = \tilde{x}$, $v = \dot{\tilde{x}}$ and $z = [e^\top \; v^\top]^\top$, the error system becomes

$$\dot{z} = Az, \qquad A = \begin{bmatrix} 0 & I \\ -\Lambda^{-1}K & -\Lambda^{-1}D \end{bmatrix}. \tag{76}$$

The matrix Λ [kg] is the Cartesian inertia, K [$\mathrm{N\,m^{-1}}$] and D [$\mathrm{N\,s\,m^{-1}}$] are the frozen stiffness and damping matrices, and A is the corresponding state matrix. The mechanical energy $V_0 = \frac{1}{2}v^\top \Lambda v + \frac{1}{2}e^\top K e$ satisfies only $\dot{V}_0 = -v^\top D v \le 0$. A strict decay rate in both e and v is obtained by adding a position–velocity cross-term:

$$V(z) = \tfrac{1}{2}v^\top \Lambda v + \tfrac{1}{2}e^\top K e + \beta e^\top \Lambda v = \tfrac{1}{2} z^\top P z, \qquad P = \begin{bmatrix} K & \beta\Lambda \\ \beta\Lambda & \Lambda \end{bmatrix}, \qquad \beta > 0. \tag{77}$$

Here β [$\mathrm{s^{-1}}$] weights the cross-term. By the Schur complement,

$$P \succ 0 \quad \Longleftrightarrow \quad K - \beta^2 \Lambda \succ 0, \tag{78}$$

for which the scalar condition $\beta^2 < \lambda_{\min}(K)/\lambda_{\max}(\Lambda)$ is sufficient. Hence, with $\underline{p} = \lambda_{\min}(P)$ and $\overline{p} = \lambda_{\max}(P)$,

$$\frac{\underline{p}}{2}\|z\|^2 \le V(z) \le \frac{\overline{p}}{2}\|z\|^2. \tag{79}$$

Theorem 1 (Exponential stability)**.** *For the nominal system* (76)*, suppose* $P \succ 0$ *and*

$$Q = \begin{bmatrix} 2\beta K & \beta D \\ \beta D & 2(D - \beta\Lambda) \end{bmatrix} \succ 0. \tag{80}$$

Then the origin $z = 0$ *is globally exponentially stable. In particular, with* $\underline{q} = \lambda_{\min}(Q)$*,*

$$\dot{V} \le -\alpha V, \qquad \alpha = \frac{\underline{q}}{\overline{p}}, \qquad \|z(t)\| \le \sqrt{\frac{\overline{p}}{\underline{p}}}\, e^{-\alpha t/2}\|z(0)\|. \tag{81}$$

Proof. Using $\Lambda\dot{v} = -Dv - Ke$, differentiation of (77) gives

$$\begin{aligned} \dot{V} &= v^\top \Lambda \dot{v} + e^\top K v + \beta v^\top \Lambda v + \beta e^\top \Lambda \dot{v} \\ &= -v^\top (D - \beta\Lambda) v - \beta e^\top K e - \beta e^\top D v = -\tfrac{1}{2} z^\top Q z. \end{aligned} \tag{82}$$

The cancellation of $-v^\top Ke + e^\top Kv$ removes the conservative spring-power exchange. From $Q \succ 0$ and (79),

$$\dot{V} \le -\frac{\underline{q}}{2}\|z\|^2 \le -\frac{\underline{q}}{\overline{p}} V = -\alpha V.$$

Integration gives $V(t) \le e^{-\alpha t} V(0)$, and substitution of the upper and lower quadratic bounds in (79) yields (81). □

Condition (80) can be checked directly from the learned gains and Cartesian inertia at each phase. Its Schur-complement form,

$$D - \beta\Lambda - \frac{\beta}{4} D K^{-1} D \succ 0, \tag{83}$$

also shows why $D - \beta\Lambda \succ 0$ alone is insufficient: the mixed term $-\beta e^\top Dv$ must also be dominated. The computable rate $\alpha = \lambda_{\min}(Q)/\lambda_{\max}(P)$ expresses the joint influence of inertia, stiffness, damping and the selected cross-term coefficient.

Time-varying gains and moving-reference perturbations. For $K = K(\phi)$, $D = D(\phi)$ and an aggregated Cartesian perturbation δ containing external contact input, imperfect model compensation and uncompensated reference acceleration, the derivative becomes

$$\dot{V} = -\tfrac{1}{2} z^\top Q(\phi) z + \tfrac{1}{2} e^\top \dot{K}(\phi) e + (v + \beta e)^\top \delta, \qquad \dot{K}(\phi) = K'(\phi)\dot{\phi}. \tag{84}$$

Corollary 1 (Uniform exponential stability with variable gains)**.** *Suppose* $P(\phi) \succ 0$ *and* $Q(\phi) \succ 0$ *uniformly for* $\phi \in [0, 1]$*, and define*

$$q_* = \inf_{\phi\in[0,1]} \lambda_{\min}\big(Q(\phi)\big), \qquad \kappa = \sup_{\phi\in[0,1]} \left\|\dot{K}(\phi)\right\|_2, \qquad \mu = q_* - \kappa. \tag{85}$$

If $\mu > 0$*, the unforced time-varying system is uniformly exponentially stable. Since* $\left\|\dot{K}(\phi)\right\|_2 \le \|K'(\phi)\|_2 |\dot{\phi}|$*, a sufficient phase-rate condition is*

$$|\dot{\phi}| < \frac{q_*}{\sup_{\phi\in[0,1]} \left\|K'(\phi)\right\|_2}. \tag{86}$$

Proof. With $\delta = 0$, equations (84) and (85) give

$$\dot{V} \leq -\tfrac{1}{2}\big(q_* - \kappa\big)\,\|z\|^2 = -\tfrac{1}{2}\mu\,\|z\|^2 .$$

Uniform positive lower and upper bounds on $P(\phi)$ imply exponential decay by the comparison argument used in Theorem 1. Condition (86) ensures $\kappa < q_*$ and therefore $\mu > 0$. □

The tank gate (75) reduces $\dot{\phi}$ as usable energy decreases and therefore reduces $\dot{K}$ and the moving-reference power demand. The torque projection (73) independently enforces the joint-port power constraint. Passivity follows from the energy accounting, whereas uniform exponential stability additionally requires the matrix margin (85); the two properties are related through phase gating but are not identical claims.

Proposition 6 (Practical exponential stability under bounded perturbations)**.** *Let the conditions of Corollary 1 hold, define* $g_\beta = \sqrt{1+\beta^2}$*, and suppose that* $\sup_{s\geq 0}\|\delta(s)\| < \infty$*. Then*

$$\dot{V} \leq -\frac{\mu}{4}\,\|z\|^2 + \frac{g_\beta^2}{\mu}\,\|\delta\|^2 \leq -aV + \frac{g_\beta^2}{\mu}\,\|\delta\|^2 , \qquad a = \frac{\mu}{2\overline{p}}, \tag{87}$$

where $\overline{p} = \sup_\phi \lambda_{\max}(P(\phi))$*. Integration gives*

$$V(t) \leq e^{-at}V(0) + \frac{g_\beta^2}{a\mu}\big(1 - e^{-at}\big)\sup_{0\leq s\leq t}\|\delta(s)\|^2 . \tag{88}$$

Accordingly, the state converges exponentially to a residual set whose radius is proportional to $\sup\|\delta\|$*; when* $\delta = 0$*, this set collapses to the origin.*

Proof. The perturbation term in (84) satisfies $|(v+\beta e)^\top \delta| \leq g_\beta\,\|z\|\,\|\delta\|$. Young's inequality with the uniform margin μ yields (87); integration of the scalar differential inequality gives (88). □

Experimental interpretation. Theorem 1 provides the frozen-gain nominal result, Corollary 1 states the uniform margin required while the learned gains evolve with phase, and Proposition 6 accounts for bounded workpiece motion, contact forces and compensation error. Figure 9 reports the corresponding empirical diagnostics. The logged cut-depth error and storage remain bounded in the plotted episodes. The fitted decay of the full Cartesian error is interpreted cautiously because low-precision axes are intentionally assigned low stiffness and therefore have smaller contraction margins.

Implementation status. The live tank state determines the torque-level budget in (73), while phase progression is gated by (75). The positive-power perturbation validation in Fig. 8 and Table 6 evaluate the implemented storage and torque-power constraints. Equations (80)–(88) separately state nominal, uniform time-varying and perturbed sufficient conditions. The runtime does not calculate $P(\phi)$, $Q(\phi)$, q_* or μ; the plots therefore diagnose boundedness but do not verify the matrix conditions of the theorems.

6 Material interaction model

In simulation the controller tracked position well, yet the contact force at first failed to match the demonstrated force. The stock simulated *material model* presents both the scalpel blade and the workpiece as rigid box geometries; hence, overlapping boxes in a rigid-contact simulator produce solver-dependent *collision forces*, unlike the physical nature of the real cutting force. Their magnitude is set by contact penetration, time step and solver parameters, and their direction by the instantaneous collision normal. A controller cannot make such a contact respect a demonstrated cutting law, because the measured "force" is a geometric contact artefact rather than a material-specific *fracture force*. Figure 10 shows this failure mode in a matched comparison: the same controller, trajectory, gains and cork workpiece are used, and only the contact model is toggled.

Proposition 7 (Why box-on-box contact is not a cutting model)**.** *Let the desired cutting interaction be a material law $F_d(\phi) = \bar{F}(d(\phi))\, r(\phi)$ with bounded magnitude $0 \le \bar{F}(d) \le F_{\text{mat}}^{\text{cut}}$ and unit reaction direction $r(\phi)$ learned from the demonstration. A rigid box-on-box MuJoCo contact generates instead*

$$F_{\text{box}} = F_{\text{solver}}(q, \dot{q}, h_{\text{sim}}, \texttt{solref}, \texttt{solimp}, \mathcal{C}(q), \mu_c), \tag{89}$$

where h_{sim} is the simulator integration step, $\mathcal{C}(q)$ is the active contact set and μ_c denotes the friction parameters. In general there exists no depth-only map $\mathcal{F} : \mathbb{R}_{\ge 0} \to \mathbb{R}^3$ that is invariant to these numerical and geometric quantities and satisfies simultaneously $F_{\text{box}} = \mathcal{F}(d) = \bar{F}(d) r(\phi)$ along the demonstrated cut.

Proof. Assume such an invariant $\mathcal{F}(d)$ exists. Consider two simulator configurations with the same blade pose and hence the same penetration d, but with distinct solver settings $\vartheta_1 = (h_{\text{sim},1}, \texttt{solref}_1, \texttt{solimp}_1)$ and $\vartheta_2 = (h_{\text{sim},2}, \texttt{solref}_2, \texttt{solimp}_2)$. The normal reaction in a penalty or constraint-regularised contact has the generic dependence

$$F_{n,i} = \Psi(d, \dot{d}; \vartheta_i), \qquad i \in \{1, 2\}, \tag{90}$$

and changing the effective contact stiffness, damping or discrete step gives $\Psi(d, \dot{d}; \vartheta_1) \neq \Psi(d, \dot{d}; \vartheta_2)$ for some $(d, \dot{d})$. Thus identical d produces different force magnitudes, which contradicts the assumed depth-only invariance.

Independently, the rigid-contact force satisfies the friction-cone restriction

$$F_{\text{box}} = F_n n_c + F_t, \qquad F_n \ge 0, \quad n_c^\top F_t = 0, \quad \|F_t\| \le \mu_c F_n, \tag{91}$$

where n_c is a normal selected by the active box feature. At a face or edge transition, $\mathcal{C}(q)$ and n_c can change while d remains continuous. The resulting force direction is therefore determined by collision geometry and need not equal the demonstrated $r(\phi)$. Since equality with the cutting law requires both

$$\|F_{\text{box}}\| = \bar{F}(d) \quad \text{and} \quad \frac{F_{\text{box}}}{\|F_{\text{box}}\|} = r(\phi),$$

failure of either condition disproves $F_{\text{box}} = \bar{F}(d) r(\phi)$. Hence the rigid collision map cannot supply the missing material-specific constitutive relation. □

The rigid contact is replaced with a *per-material cutting-force law*. Physically, cutting force rises elastically with penetration and then *plateaus* at the material's cutting strength (the blade fractures/cuts the material). Let $p_b(t) \in \mathbb{R}^3$ be the blade-tip position, $p_s(t) \in \mathbb{R}^3$ a point on the live workpiece surface, and $n_s(t) \in \mathbb{R}^3$, $\|n_s\| = 1$, its outward normal. The instantaneous penetration is

$$d(t) = \Big[-n_s(t)^\top \big(p_b(t) - p_s(t) \big) \Big]_+, \qquad [\xi]_+ = \max(\xi, 0). \tag{92}$$

Using the current $p_s(t)$ rather than its initial value preserves the penetration definition for a moving workpiece. Let $\hat{F}_d(\phi) = F_d(\phi) / (\|F_d(\phi)\| + \varepsilon_F)$ denote the regularised demonstrated force direction. The reaction applied to the blade is

$$\boxed{\begin{aligned} f_{\text{mat}}(d; \theta_{\text{mat}}) &= \min\big(k_{\text{mat}} d, F_{\text{mat}}^{\text{cut}}\big), \\ F_{\text{react}}(d, \phi) &= -f_{\text{mat}}(d; \theta_{\text{mat}})\, \hat{F}_d(\phi), \\ \theta_{\text{mat}} &= \big(k_{\text{mat}}, F_{\text{mat}}^{\text{cut}}\big), \qquad d_c = \frac{F_{\text{mat}}^{\text{cut}}}{k_{\text{mat}}}. \end{aligned}} \tag{93}$$

Here k_{mat} [N m^{-1}] is the pre-fracture loading stiffness, $F_{\text{mat}}^{\text{cut}}$ [N] is the cutting-force plateau, and d_c [m] is the transition depth. Equivalently,

$$f_{\text{mat}}(d) = \begin{cases} 0, & d = 0, \\ k_{\text{mat}} d, & 0 < d < d_c, \\ F_{\text{mat}}^{\text{cut}}, & d \ge d_c. \end{cases} \tag{94}$$

The first branch represents separation, the linear branch represents material loading before fracture, and the plateau represents continued cutting after the force threshold has been reached.

Proposition 8 (Boundedness and regularity of the identified cutting law)**.** *For $k_{\mathrm{mat}} > 0$ and $F_{\mathrm{mat}}^{\mathrm{cut}} > 0$, the scalar law* (94) *is non-negative, monotone non-decreasing, globally bounded by $F_{\mathrm{mat}}^{\mathrm{cut}}$ and globally Lipschitz on $\mathbb{R}_{\geq 0}$ with constant k_{mat}. Its scalar work primitive is*

$$U_{\mathrm{mat}}(d) = \int_0^d f_{\mathrm{mat}}(s)\, ds = \begin{cases} \frac{1}{2} k_{\mathrm{mat}} d^2, & 0 \leq d \leq d_c, \\ F_{\mathrm{mat}}^{\mathrm{cut}} d - \dfrac{(F_{\mathrm{mat}}^{\mathrm{cut}})^2}{2k_{\mathrm{mat}}}, & d > d_c. \end{cases} \tag{95}$$

Proof. Both arguments of the minimum in (93) are non-negative and non-decreasing for $d \geq 0$, which establishes non-negativity and monotonicity. The minimum is at most $F_{\mathrm{mat}}^{\mathrm{cut}}$. For any $d_1, d_2 \geq 0$,

$$|f_{\mathrm{mat}}(d_1) - f_{\mathrm{mat}}(d_2)| \leq k_{\mathrm{mat}} |d_1 - d_2|, \tag{96}$$

because saturation cannot increase the slope of the unsaturated linear map. Integration of the two branches gives (95); at $d = d_c$, both expressions equal $(F_{\mathrm{mat}}^{\mathrm{cut}})^2/(2k_{\mathrm{mat}})$, proving continuity. □

Parameter identification. For each material, valid samples satisfy $d_j > 10^{-4}\,\mathrm{m}$ and have force magnitude $y_j = \|F_j\|$. The implemented estimator solves the bounded robust nonlinear least-squares problem

$$\hat{\theta}_{\mathrm{mat}} = \arg \min_{k>0,\, F^{\mathrm{cut}}>0} \sum_{j=1}^{N_{\mathrm{mat}}} \rho_{\mathrm{softL1}}\big(y_j - \min\big(k d_j, F^{\mathrm{cut}}\big)\big)\,, \tag{97}$$

with numerical bounds $1 \leq k \leq 10^6\,\mathrm{N\,m^{-1}}$ and $1 \leq F^{\mathrm{cut}} \leq 500\,\mathrm{N}$. The initial plateau is the 75th percentile of valid force magnitudes and the initial stiffness makes that plateau occur at 0.3 times the median valid depth. If optimization fails, the initial values are retained; if fewer than three valid samples exist, the code uses $(8000\,\mathrm{N\,m^{-1}}, 50\,\mathrm{N})$. Thus the estimator is now the implemented robust fit, but its parameters are still calibrated and evaluated on demonstration-derived data rather than an independent test set.

The reaction direction used by the implementation is the negative instantaneous desired-force direction:

$$\hat{r}(\phi) = -\frac{F_d(\phi)}{\|F_d(\phi)\| + \varepsilon_F}, \tag{98}$$

with zero reaction below the numerical force threshold. Hence the force direction is replayed from the prior and is not independently identified or predicted by the material law.

The geometric box contact is disabled during the cutting experiment, F_{react} from (93) is applied to the blade, and the force sensor reports the same interaction. Parameter sensitivity is bounded by

$$|f_{\mathrm{mat}}(d; k_1, F_1) - f_{\mathrm{mat}}(d; k_2, F_2)| \leq d|k_1 - k_2| + |F_1 - F_2|, \tag{99}$$

so errors in the identified loading stiffness dominate at larger pre-fracture penetrations, whereas plateau error directly bounds the post-fracture force error.

7 Results

Figure 11 summarises the closed-loop behaviour with the identified material on both a static and a moving workpiece.

Figure 12 reports the earlier material-model replay for PVC and penoplex. Those plots assess consistency of a model identified from the same demonstrations; they are not independent evidence of force-regulation accuracy. Moreover, their moving-workpiece traces were generated under an earlier reference convention and are not used to support the fixed-world reference ablation of Section 4.5.

Material	$F_{\text{mat}}^{\text{cut}}$ (N)	k_{mat} (N/m)	contact maintained	force RMSE (N)
penoplex (foam)	63	9,104	100%	1.8
cork	59	9,917	100%	2.5
PVC (hardest)	86	13,855	100%	2.6

Table 7: Cutting parameters identified per material, and the resulting force tracking. The materials are clearly distinct (PVC is stiffest and strongest, consistent with its hardness), and the force matches the demonstrated level to within $\approx 2\,\text{N}$ — an approximate physical fit, not an exact replay.

- **The task performance is attributable to the learned variance structure** (Table 3, Fig. 3): the identified precision has the lowest mean cut-depth error: 6.0%, 33.8% and 38.6% below the permuted, inverted and direct-force comparators, respectively. Paired significance is $p = 0.001$, 0.001 and 0.002; absolute error remains highly dispersed.
- **Material-model replay** (Table 7, Fig. 11): the identified reaction reproduces the demonstrated force scale in the calibration setting. Because identification and assessment use the same demonstration source, this is a consistency check rather than an independent force-control validation.
- **Position tracking**: deterministic nominal cut-depth RMSE is 1.781–2.370 mm in the four-case cork comparison. Full Cartesian error is larger, and the 120 N holdback produces 26.533–114.292 mm RMSE; no high-precision Cartesian tracking claim is made.
- **Passivity** (Fig. 8, Table 6): under positive joint-power perturbation, the controller without tank projection violates the available energy budget for 69.8% of the samples (+24,586 W maximum margin), while both tanked controllers satisfy $\dot{q}^\top \tau_{\text{safe}} \leq P_{\max}$ for the whole run. With the deliberately small 1 J stress-test tank, the phase-gated case halts at $\phi = 0.024$ rather than completing.
- **Stability diagnostics** (Fig. 9): nominal static and moving cuts remain bounded and complete the task; cut-depth error stays small. The exponential fit is reported as a diagnostic, while the full Cartesian error norm is interpreted carefully because low-precision axes are intentionally released.
- **Run monitoring**: Figure 13 shows the monitored signals for static and moving workpieces, with the validation holdback disabled and enabled: position, material reaction, stiffness, tracking error and tank energy, each in its own units.

8 Novelty and contributions

1. **One progress-registered material-conditioned GMM/GMR.** All 96 demonstrations train one structured probabilistic prior conditioned on phase and categorical material contrasts (49). Spatial-progress registration (48) removes the sample-time misregistration that curved the legacy GMR mean, while phase–material interactions retain material-specific affine geometry and nonlinear interaction profiles. The straightness audit in Table 4 is performed on the saved learned prior before robot execution.
2. **Gains learned from demonstration precision, via an online QP.** Stiffness *is* the demonstration precision (22), computed each step by the QP (23) that blends force-consistency with precision-stiffness under force-saturation constraints. The controlled comparison (Table 3) shows the *specific* learned variance structure — not generic adaptivity — is what carries the skill.

3. **State-driven task parametrisation.** A phase variable advanced by matching the robot's measured state to the prior (67), rather than by a clock, giving speed-adaptability and disturbance awareness, and unifying with the stability mechanism.

4. **Torque-port passivity under variable gains and a moving target.** The live energy tank sets the torque-level power budget (73); Fig. 8 demonstrates that this rejects positive-power perturbations. A cross-term Lyapunov proof (Theorem 1) gives the exponential-contraction condition for the regulated error dynamics, and the tank provides the mechanism that bounds phase and gain rates.

5. **An empirical principle for contact-skill transfer.** Position is the reproducible prior, force is not; the controller therefore regulates position and treats force as a constraint, while a per-material identified model supplies the interaction force.

9 Assumptions, scope and limitations

- The contact *force* matches because the material model is identified to the demonstration (system identification); the *controller* achieves the position/velocity tracking. These two roles are reported separately, and the force result is *not* presented as controller force-tracking.

- All experiments are in simulation; the material parameters are identified from the real material's measured demonstration forces. Real-robot validation is future work.

- Lateral (Y) tracking lags the large demonstrated lateral swing; the task-critical cut-depth is tight. Reducing the lateral error is a phase-speed / parametrisation question, separate from the force model.

- The energy-tank passivity guarantee is implemented and validated at the joint torque port. The reconstructed Cartesian residual is retained as a useful diagnostic, but it is not the primary proof artifact until the force frame/sign, finite-difference velocity estimate and Cartesian storage model are matched more precisely.

- The exponential-stability proof applies under regulation assumptions (constant or sufficiently slowly varying gains, passive environment, bounded reference acceleration). The current simulations support bounded stable cutting; a stronger empirical exponential-convergence claim should be made on the regulated cut-depth or precision-weighted error, not on the full Cartesian norm that includes intentionally released axes.

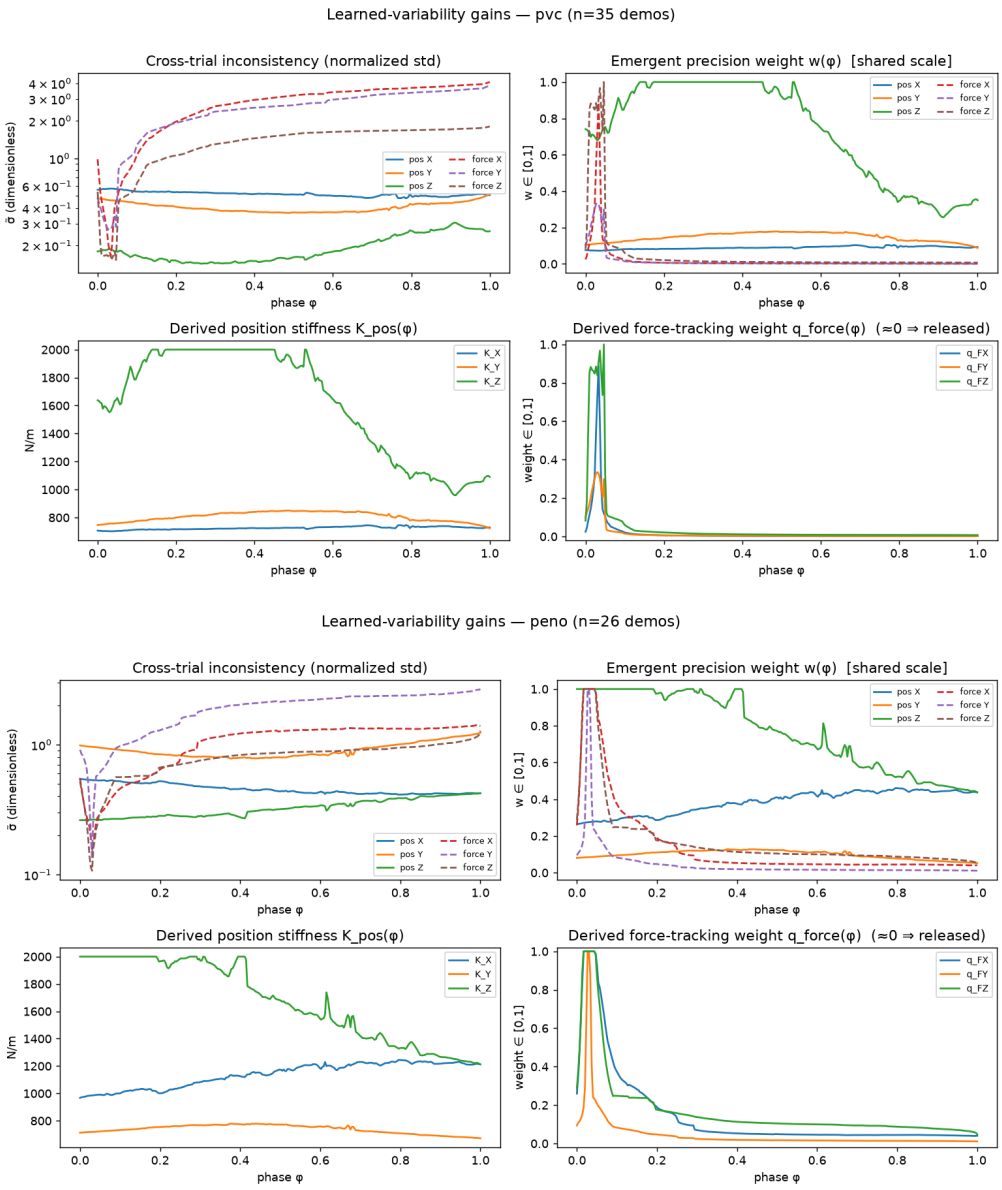


Figure 2: Learned gains for PVC (top) and penoplex (bottom), produced by the same precision rule (22) on each material's own demonstrations. The structure is consistent with cork (Fig. 1) — the depth axis is tracked stiffly and force is released — while the stiffness magnitudes and profiles differ per material, so the controller expresses material-specific behaviour without any manual retuning.

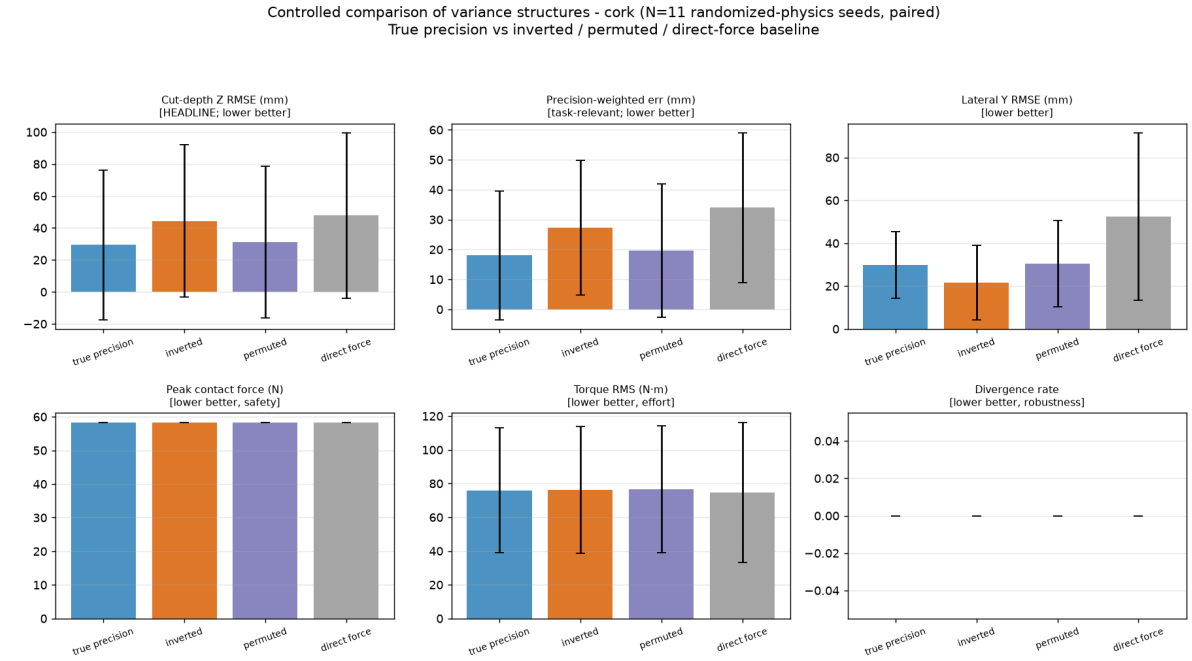


Figure 3: Controlled comparison of gain assignments on cork ($N_{\text{seed}} = 11$ randomised-physics seeds, paired). Only the identified precision attains low cut-depth error; inverting or permuting the learned structure degrades it. Bars denote mean±std.

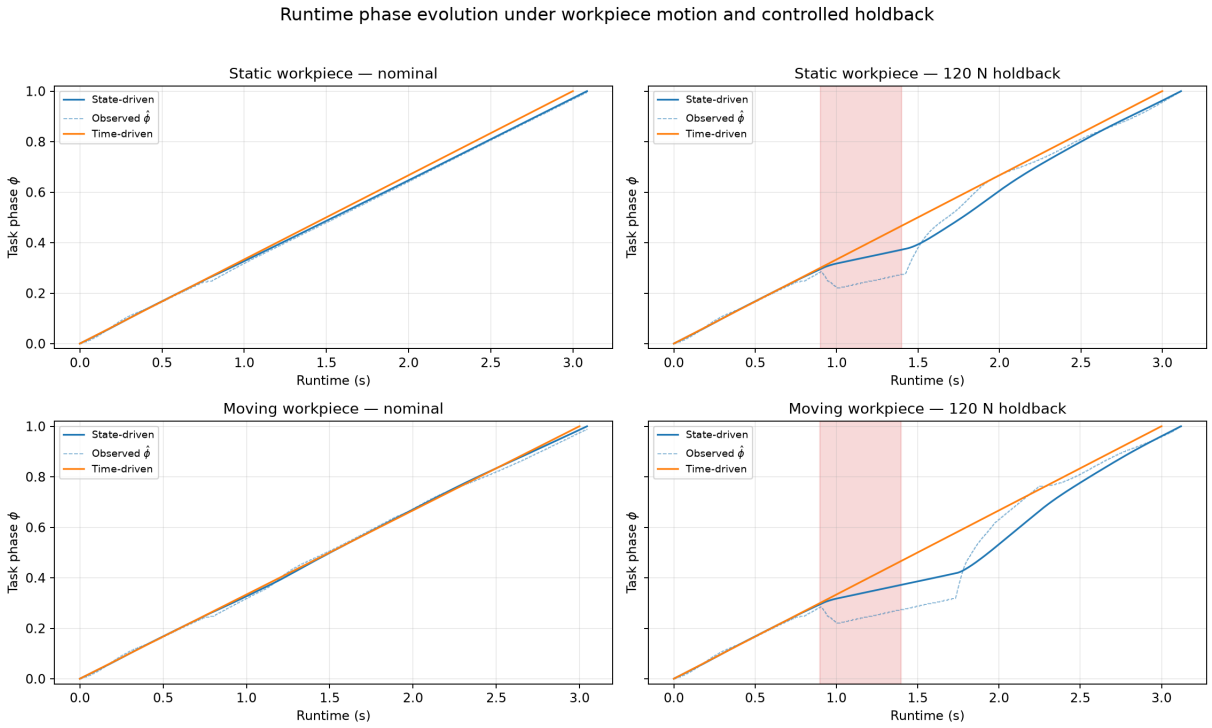


Figure 4: Representative runtime phase evolution for static and moving workpieces, with and without the controlled holdback. The red interval is the physical perturbation (70). In nominal runs the two laws remain close. Under holdback, the measured projection $\hat{\phi}$ falls behind the commanded phase and the state-driven law retards progress, whereas the time-driven phase continues at its clock rate.

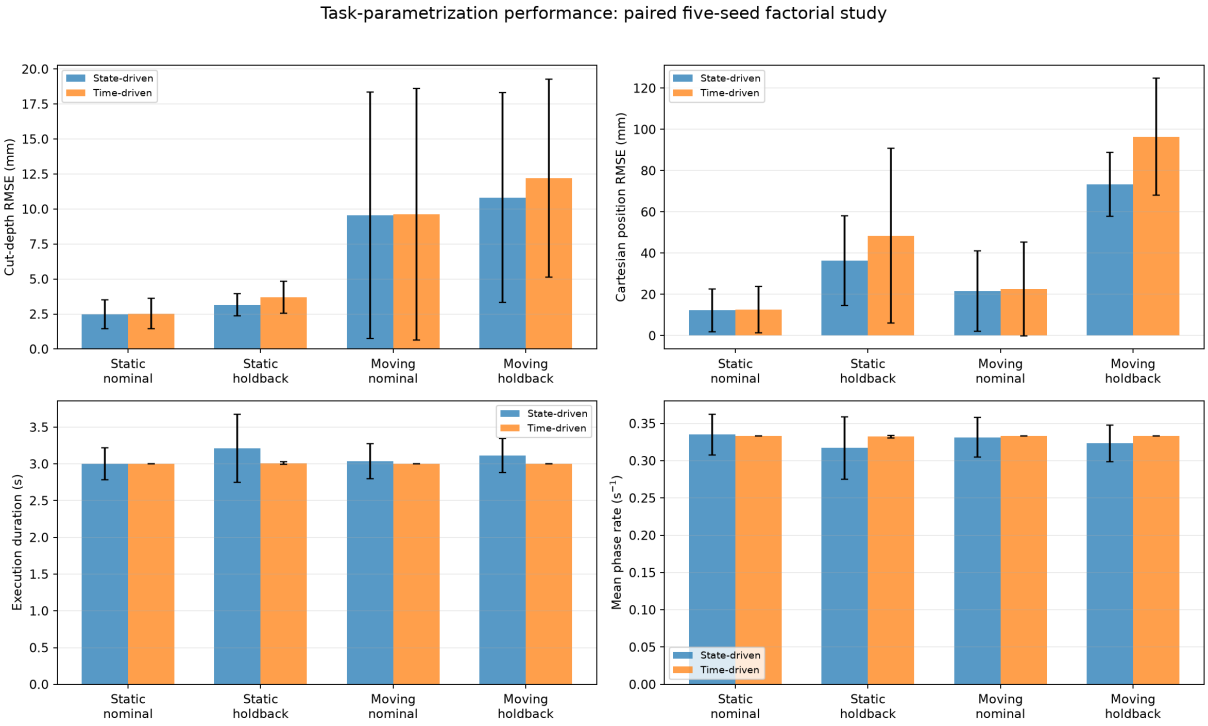


Figure 5: Performance over the complete factorial study. Error bars show one standard deviation across paired randomised-physics seeds. State-dependent timing trades execution time for synchronisation when measured progress lags; its rate can also exceed the clock rate when the state projection is ahead.

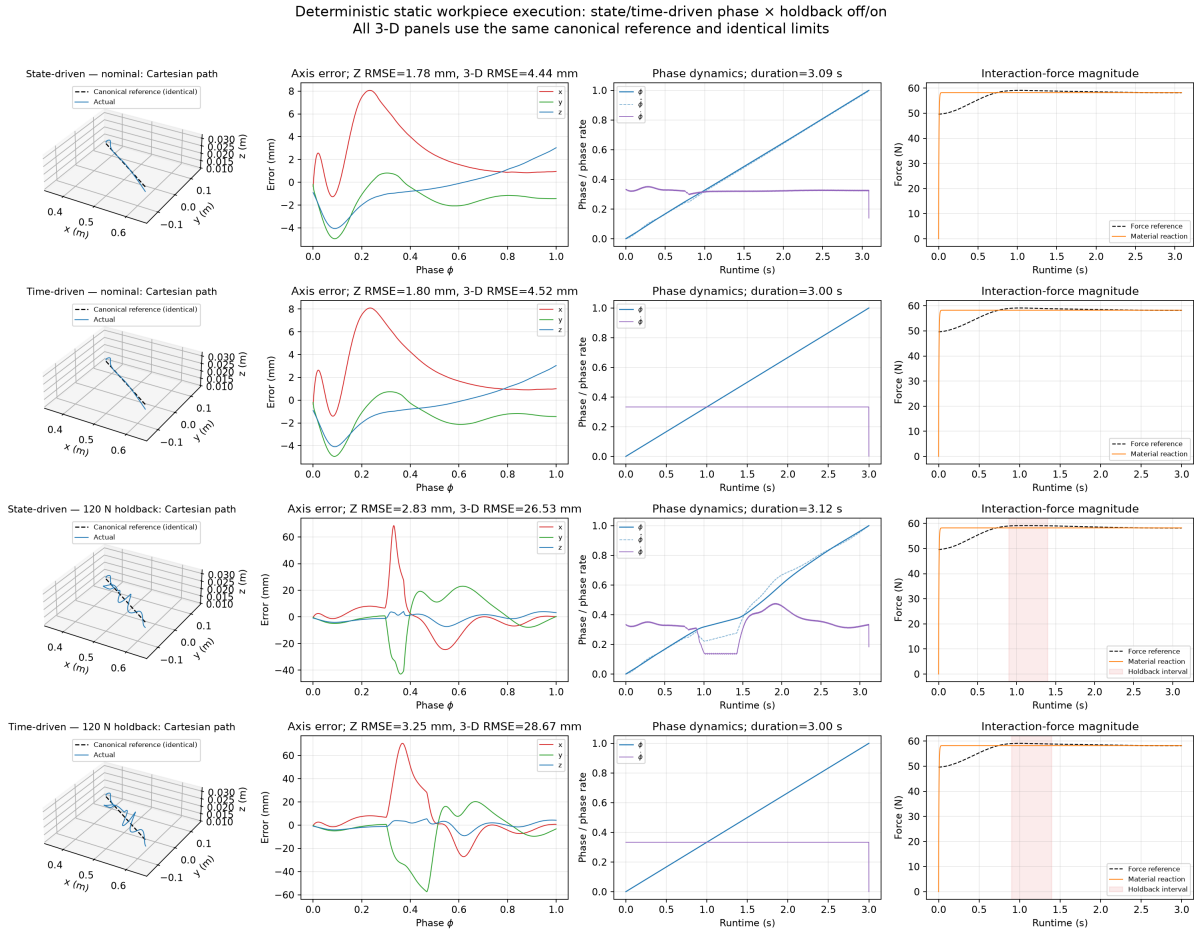


Figure 6: Deterministic static-workpiece execution without domain randomisation. Rows cross state-driven/time-driven phase with holdback disabled/enabled; columns show the Cartesian path, axis errors, phase dynamics and interaction force. Every 3-D panel uses the same canonical geometric reference and identical axis limits; only the executed trajectory changes. Nominal cut-depth RMSE is 1.781 and 1.798 mm for state- and time-driven execution, respectively. Under holdback it is 2.831 and 3.248 mm. The corresponding full Cartesian RMSE increases to 26.533 and 28.669 mm. Thus this 120 N test demonstrates constraint handling and phase response under a severe disturbance, not high-accuracy rejection of that disturbance.

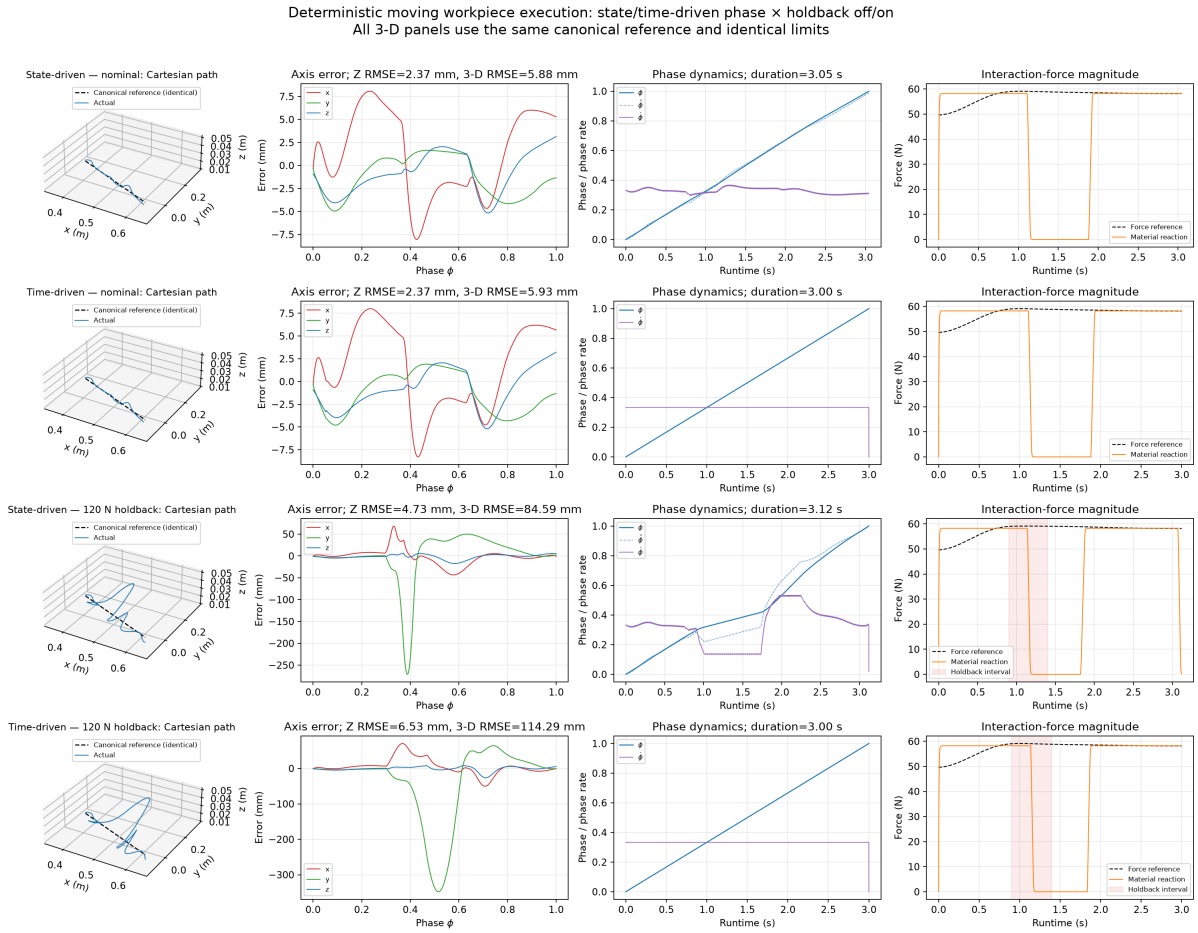


Figure 7: Deterministic moving-workpiece execution under the same four phase/holdback conditions as Fig. 6. Nominal cut-depth RMSE is 2.369 mm for state-driven timing and 2.370 mm for time-driven timing. Under holdback the corresponding values are 4.733 and 6.530 mm, and full Cartesian RMSE is 84.593 and 114.292 mm. Workpiece motion changes only the environment geometry: it does not translate the reference. Loss of contact can consequently occur when the material moves away from the fixed commanded path.

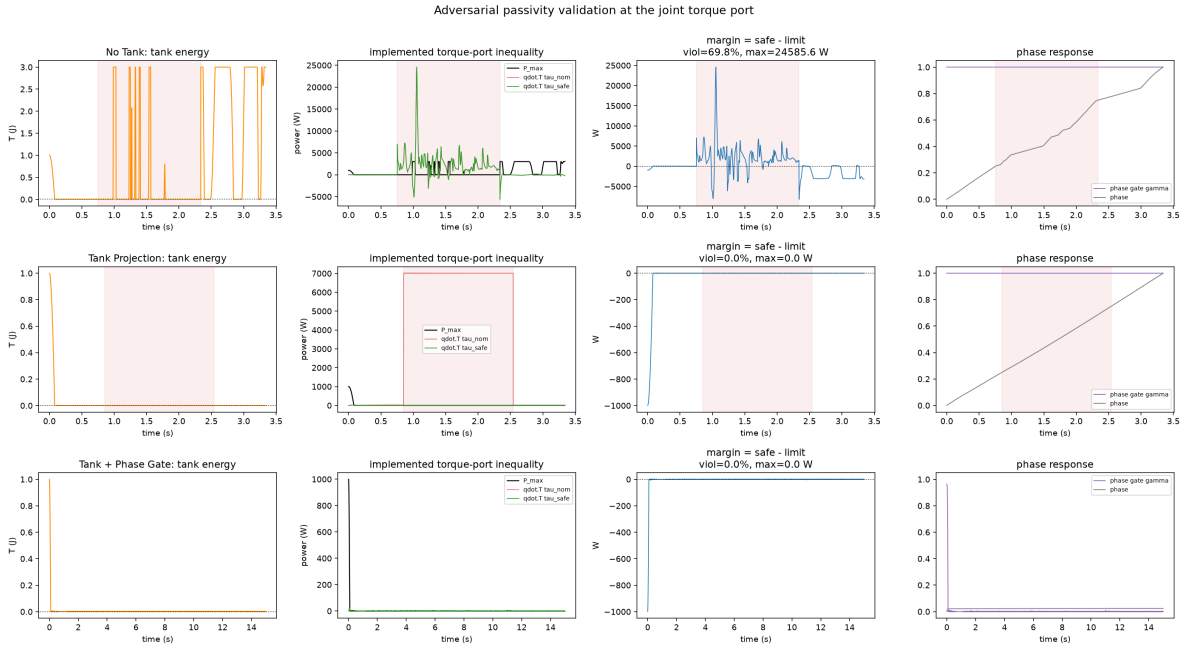


Figure 8: Positive-power perturbation validation on cork. A known positive joint-power burst is injected before the torque projection. Without projection (top row), $\dot{q}^\top \tau$ exceeds the counterfactual budget for 69.8% of samples and reaches a maximum margin of +24,586 W. With torque projection (middle and bottom), the implemented inequality $\dot{q}^\top \tau_{\text{safe}} - P_{\max} \leq 0$ has 0.0% logged violations. With the deliberately small 1 J validation tank, phase gating stops progress at $\phi = 0.024$ after storage depletion; safety intervention therefore does not imply task completion.

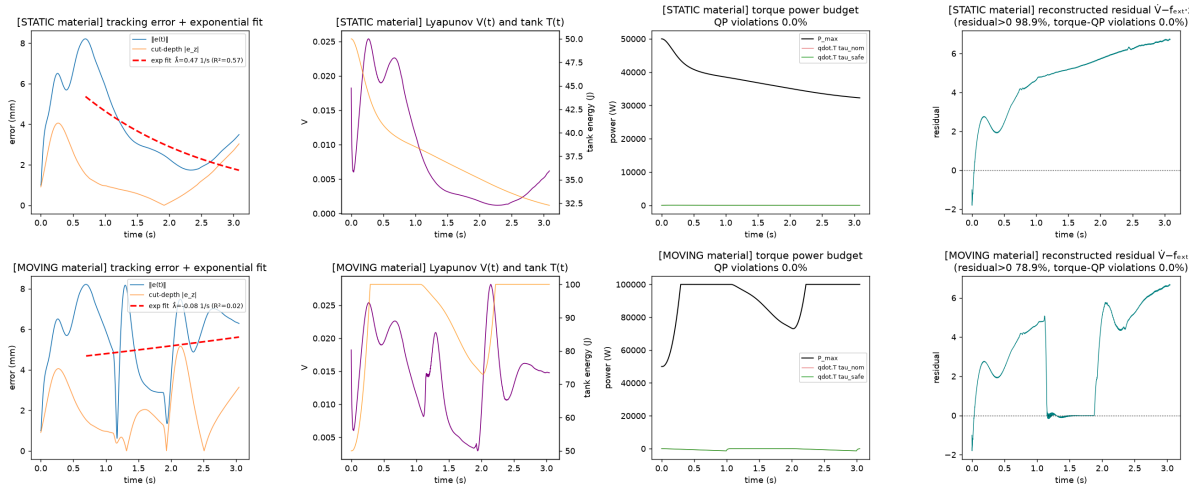


Figure 9: Nominal stability and passivity diagnostics on cork. The left column shows tracking error and cut-depth error; the middle column shows the Lyapunov diagnostic and tank energy; the third column shows the implemented torque power budget $P_{\max}$ together with nominal and projected torque power; the right column shows the reconstructed Cartesian residual, used as a diagnostic rather than as the primary passivity proof.

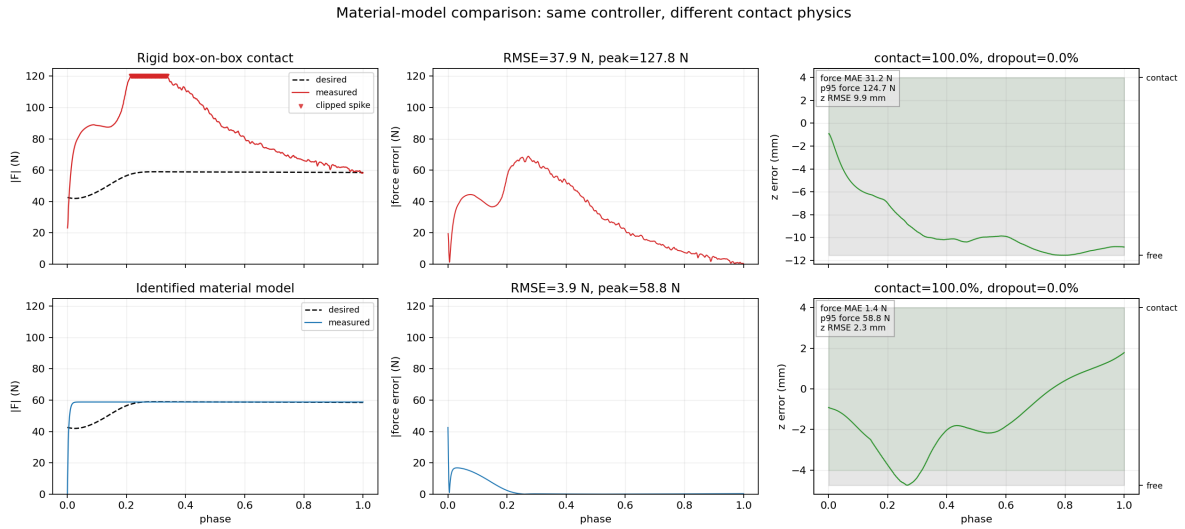


Figure 10: Why a material model is needed. This matched cork comparison uses the same controller and toggles only the contact physics. Top row: rigid box-on-box contact over-reacts and follows the solver geometry rather than the demonstrated cutting plateau (37.9 N force RMSE, 127.8 N peak, 9.9 mm depth RMSE). Bottom row: the identified cutting-force law produces the bounded material plateau and tracks the demonstrated force (3.9 N force RMSE, 58.8 N peak, 2.3 mm depth RMSE).

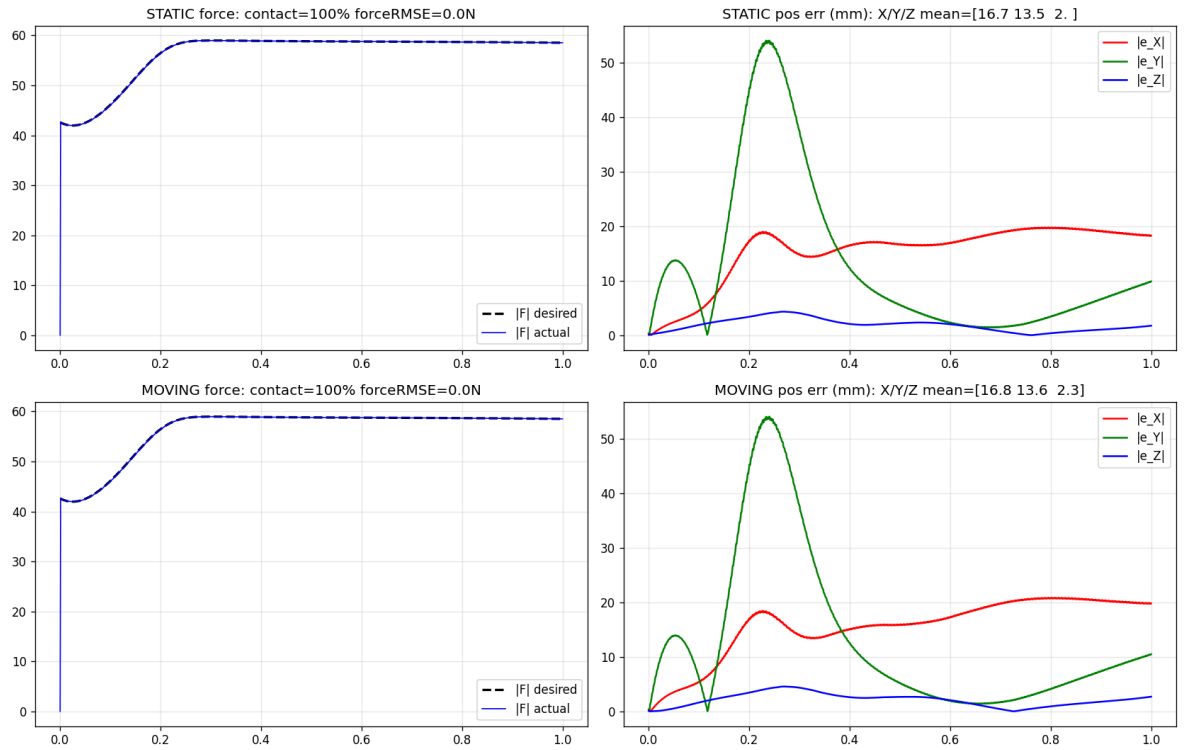


Figure 11: Cutting with the identified material model on **cork**. Left column: the measured contact-force magnitude (blue) follows the desired (dashed) for the whole cut, on static (top) and moving (bottom) workpieces. Right column: per-axis position error — the cut-depth (Z) error stays ≈ 2 mm; the lateral (Y) error peaks during the large demonstrated lateral swing.

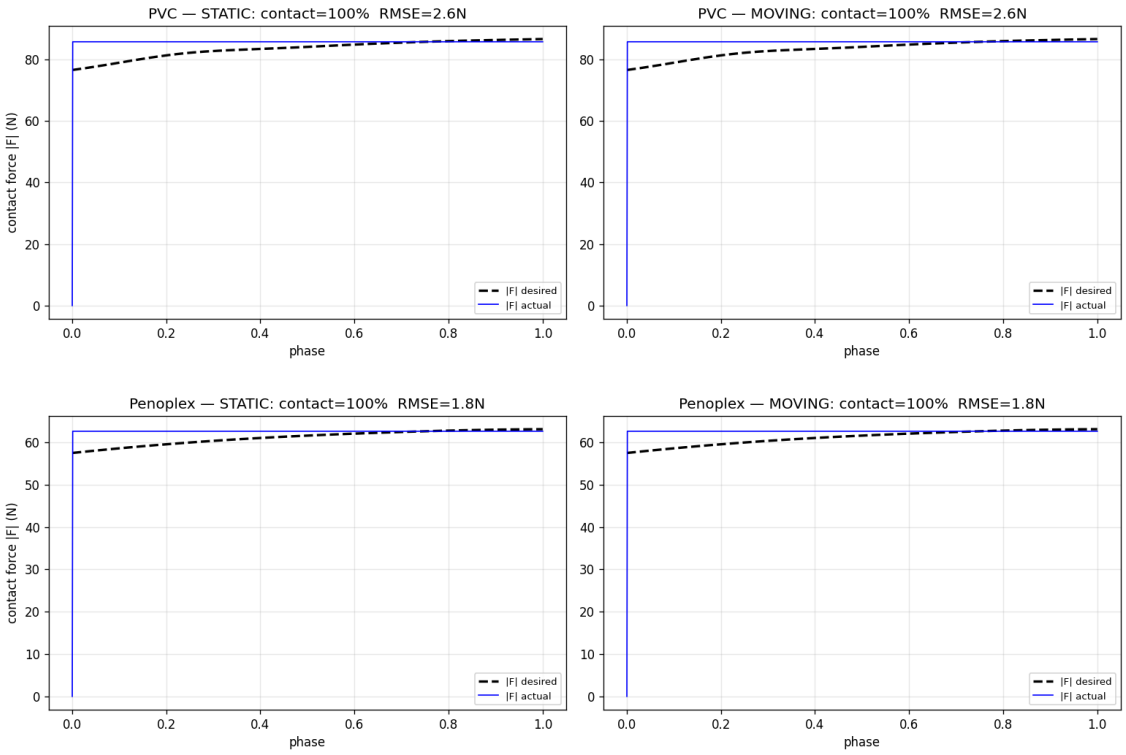


Figure 12: Force tracking on **PVC** (top) and **penoplex** (bottom), static (left) and moving (right). The measured force (blue) follows each material's demonstrated level (dashed) to $\approx 2\,\mathrm{N}$ RMSE with 100% contact. PVC plateaus highest ($\sim 86\,\mathrm{N}$, hardest material), penoplex lowest ($\sim 63\,\mathrm{N}$), mirroring the per-material cutting parameters of Table 7.

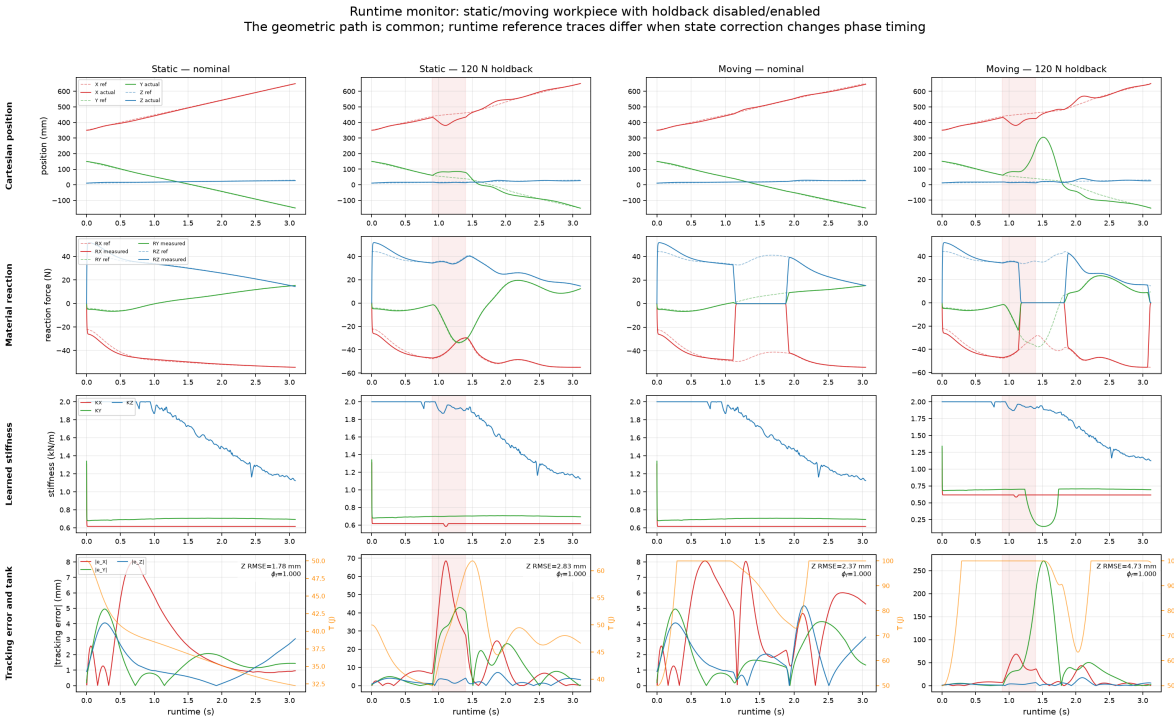


Figure 13: Runtime monitor reconstructed from four state-driven cork episodes: static nominal, static with 120 N holdback, moving nominal, and moving with holdback (columns from left to right). Rows show Cartesian position, axis-resolved material reaction, learned stiffness, and absolute tracking error with tank energy. Red shading identifies the validation-force interval $[0.9, 1.4]$ s; the plotted reaction channel remains the identified material reaction and does not relabel the separately applied holdback as contact force. The desired world-frame path is common to all columns, although its time-indexed samples differ when state-driven phase timing changes. Representative cut-depth RMSE is 1.781, 2.831, 2.369 and 4.733 mm, respectively. The holdback columns exhibit large Cartesian displacement and must be interpreted as stress tests rather than precision-tracking results.